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Machine learning techniques in bankruptcy prediction: A systematic literature review

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A R T I C L E   I N F O

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A B S T R A C T

The main objective of this systematic literature review is to unveil the prevailing trend of employing cutting-edge models for bankruptcy prediction for a period spanning from 2012 to mid-2023. Employing the PRISMA method, we reviewed 207 empirical studies on bankruptcy prediction. Prior extensive research has shown that the integration of more advanced techniques, such as hybrid model, enhances prediction accuracy and robustness, leading to more reliable bankruptcy forecasts. While financial ratios have traditionally played a central role in bankruptcy prediction models, this review places emphasis on the significance of incorporating non-financial ratios. Non-financial ratios capture qualitative and intangible factors such as management competence, corporate governance practices and market reputation. The inclusion of these non-financial ratios, alongside financial ratios in hybrid models, enables a more comprehensive evaluation of a firm's financial health and improves the accuracy of bankruptcy predictions. The review also addresses the challenges and limitations associated with hybrid models and the incorporation of non-financial ratios. Our research shows that there is a current trend toward the development of hybrid models that combine multiple methodologies and variables to improve bankruptcy prediction accuracy. Researchers are actively addressing the challenge of imbalanced datasets in bankruptcy prediction by exploring and developing specialized techniques for handling such data. Moreover, during the evaluation of bankruptcy prediction models, it is essential to consider a range of metrics, including sensitivity and specificity, along with other relevant metrics, to obtain a comprehensive assessment of model performance.

1. Introduction

In today's dynamic and highly competitive business landscape, the accurate prediction of financial distress and bankruptcy has emerged as a critical concern for various stakeholders, including investors, creditors, regulators, and managers. The ramifications of bankruptcy are far-reaching, leading to significant economic losses, job reductions, and financial instability. Therefore, the ability to predict bankruptcy accurately and timely is of paramount importance for effective financial decision-making, risk management, and the overall health of the economy (Shi & Li, 2019a).

Bankruptcy, often considered a disruptive event, can be triggered by a multitude of factors. Financial mismanagement, deteriorating market conditions, regulatory changes, and unforeseen external shocks (Kirkos, 2015) are just a few examples of the complex interplay that can drive a company toward insolvency. The ability to identify the warning signs and anticipate such adverse events well in advance can empower stakeholders to take proactive measures, mitigate risks, and potentially avert bankruptcy (Zhu, 2016).

Traditionally, bankruptcy prediction had relied on conventional financial ratios and quantitative analysis, which often struggled to capture the intricacies and nonlinear relationships inherent in complex financial data. However, the rapid advancements in artificial intelligence (AI) and machine learning (ML) techniques have opened new avenues for bankruptcy prediction, providing more sophisticated tools to analyze vast amounts of data, uncover hidden patterns, and improve the accuracy of predictions.

The application of AI and ML algorithms for bankruptcy prediction has gained significant academic attention. The pertinent literature has well acknowledged the wide range of methodologies employed (Ahmed et al., 2022; Ciampi et al., 2021; Lahmiri et al., 2020; Lommers et al., 2021; Shi & Li, 2019a; Devi & Radhika, 2019; Devi & Radhika, 2018; Kirkos, 2015; du Jardin & Séverin, 2012). Furthermore, recent advancements have led to the development of new predictive models

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(Zhao et al., 2023; Aly et al., 2022; Brenes et al., 2022; Kuang et al., 2022; Lombardo et al., 2022; Gavurova et al., 2022; Hilal et al., 2022; Kim et al., 2022; du Jardin, 2021a; Eygi Erdogan et al., 2021; Yang et al., 2021; Ye, 2021; Jandaghi et al., 2021; Park et al., 2021; Korol, 2013). These new techniques offer the potential to uncover non-obvious relationships between financial variables, detect subtle indicators of financial distress, and enhance the predictive power of bankruptcy models. By leveraging the power of computational models, researchers and practitioners aim to develop robust and reliable bankruptcy prediction models that can assist decision-makers in identifying companies at risk of insolvency and guide appropriate intervention strategies (Mishraz et al., 2021; Yang et al., 2021; Nyitrai & Virág, 2019).

Despite the growing interest and advancements in bankruptcy prediction using AI and ML, several challenges have emerged. One of the primary challenges is the availability and quality of data (Aljawazneh et al., 2021). Financial data is often complex, heterogeneous, and prone to errors, requiring careful preprocessing and feature engineering to extract meaningful insights. Additionally, the dynamic nature of financial markets demands the development of models that can adapt to changing economic conditions (Safi et al., 2022). Another significant challenge is the interpretability and explainability of AI and ML models. For example, Park & Yang (2022) studied the explainability of machine learning models for bankruptcy prediction assuming that when machine learning models have sufficient interpretability, they would have the potential to be used as effective analytical models in bankruptcy prediction.

As these models become increasingly sophisticated, they often function as "black boxes" making it difficult to understand the reasoning behind their predictions. Jones et al. (2017) explained that "black boxes" are designed to capture complex nonlinear relationships and interactions which are largely hidden in the internal mathematics of the model system. Despite the high predictability of the models, their practical usefulness and appeal could be limited if they are too difficult to implement or interpret in practice. The lack of interpretability raises concerns among stakeholders who require transparency and justification for the decision-making process, particularly in sensitive domains such as finance.

Adopting the PRISMA protocol (Page et al., 2021), this paper applies a systematic literature review process to investigate bankruptcy prediction methods employing AI and ML techniques. The selection process of the most relevant papers was carried out by searching the theme in the well-known and extensive Scopus and Web of Science databases and validating the group of articles selected as a representative sample of the literature. Scopus database covers nearly 27,950 active titles and contains title of major publishers like Springer Nature, Wiley Blackwell, IEEE, Elsevier, Taylor & Francis and more.[1] Web of Science covers nearly 254 research areas ang give the opportunity to accelerate a novel research of high quality.[2]

To the best of our knowledge, this is the first attempt to collate papers that examine bankruptcy prediction methods using AI and ML techniques, with a specific focus on hybrid models, input variables, performance metrics, and imbalanced data. Nazareth and Ramana Reddy (2023), for example, did not solely focus on bankruptcy prediction but instead examined various finance-related issues, including stock market prediction, cryptocurrency prediction, and forex prediction. In the realm of bankruptcy prediction, they analyzed a limited number of ten articles. Ahmed et al. (2022), Shi & Li (2019a), and Ciampi et al. (2021) carried out bibliometric reviews, emphasizing quantitative assessments of academic output. Kuizinienė et al. (2022) performed a comparative literature review, covering subject areas such as imbalanced data and feature selection methods but omitting exploration of hybrid and ensemble methods. Kim et al. (2020) conducted a literature review on corporate bankruptcy prediction, presenting representative methodologies and significant studies categorized by statistical approaches and machine learning techniques. Shi & Li (2019b) conducted a literature review aimed at identifying core international academic papers related to bankruptcy prediction using the Scopus database, with a primary goal of observing paper evolution from 1968 to 2017. Devi & Radhika (2018) discussed bankruptcy prediction issues in terms of selected methods and performance metrics, examining 37 articles. Alaka et al. (2018) conducted a systematic review covering the period 2010 to 2015, reviewing 49 articles related to bankruptcy prediction, with specific emphasis on methods such as Multiple Discriminant Analysis (MDA), Logistic Regression (LR), Artificial Neural Network (ANN), Support Vector Machines (SVM), Random Subspaces (RS), Case based Reasoning (CBR), Decision Trees (DT,) and Genetic Algorithms (GA). The authors also proposed the need for the study of ensemble and hybrid models. Lin et al. (2012) reviewed 130 articles related to bankruptcy prediction from 1995 to 2010, focusing on the development of state-of-the-art machine learning techniques, including hybrid and ensemble methods. The authors also highlighted the issue of imbalanced data for future research.

This systematic literature review harbors three contributions. First, this study extends prior literature by reviewing articles published up to the mid of 2023. Second, this study focuses on current trends that have not been covered by prior literature reviews, such as the exploration of hybrid models and the management of imbalanced data. Third, this study reviews articles on bankruptcy prediction employing data from both banks and companies, while previous literature reviews mostly focused on corporate bankruptcy. In sum, our paper contributes to the systematic literature review on bankruptcy prediction by extending the period of investigation, broadening the focus of articles (i.e. companies and banks) and covering trends never examined before (i.e. hybrid models and imbalanced data).

To achieve this objective, we address the following fundamental research questions:

(1) What Machine Learning (ML) and Artificial Intelligence (AI) methods have been applied in bankruptcy prediction studies?

(2) What types of input datasets, variables, and variable types are commonly used in these studies? And, how have researchers addressed the issue of imbalanced data in bankruptcy prediction, and what methods have been employed to mitigate this problem?

(3) Which performance metrics have been predominantly selected to evaluate the effectiveness of bankruptcy prediction models?

In addition to these overarching research objectives, this study contributes to the discourse by evaluating the strengths and limitations of various methods employed in bankruptcy prediction over the last decade. We believe that our study will help researchers explore novel avenues in bankruptcy prediction and facilitate stakeholders to make optimal decision-making, taking into account financial risks.

The structure of this paper is as follows: Section 2 presents the research methodology and discusses the research approach. Section 3 presents the findings of the literature review and addresses the research questions. Section 4 summarizes the main conclusions of the study.

2. Methodology

This section presents the methodology employed for data collection and the subsequent review process. The study adheres to the PRISMA methodology, as described by (Page et al., 2021). PRISMA stands for "Preferred Reporting Items for Systematic Reviews and Meta-Analyses" and is structured into four stages: Identification, Screening, Eligibility, and Inclusion. The PRISMA method has been extensively used by prior studies such as Nazareth and Ramana Reddy (2023).

[1] https://www.elsevier.com/?a=69451.

[2] https://clarivate.com/products/scientific-and-academic-research/research-discovery-and-workflow-solutions/webofscience-platform/.

2.1. Identification

The identification stage involves outlining the search strategy that we followed. In this study, we focused on “articles”, and “reviews” written in English and published between 2012 and mid-2023. We selected this specific period because previous literature review studies have covered periods before 2012 (e.g., Kirkos 2015; Lin et al., 2012). The search in Scopus and Web of Science was conducted during July 2023. Following similar literature review studies (Nazareth & Ramana Reddy, 2023; Kuizinienė et al., 2022; Shi & Li, 2019a, 2019b), we utilized the following keywords and we run the following query in Scopus and Web of Science using the operator “TITLE-ABS-KEY”:

((“Financial crisis” OR “bankruptcy” OR “business failure” OR “distress” OR “default”)) AND (“artificial intelligence” OR “machine learning”) AND (“business” OR “firm” OR “company” OR “bank”).

This search resulted in a total of 663 studies. Adopting the PRISMA methodology, we aim to ensure the transparency, accuracy, and reliability of our systematic review and *meta*-analysis. The subsequent stages of screening, eligibility, and inclusion enabled us to select the most relevant and appropriate studies to address the research objectives effectively.

2.2. Screening

The screening stage serves as a crucial step in the review process, aiming to eliminate duplicate and irrelevant articles from consideration. During this phase, an in-depth examination of the titles and abstracts of each study was conducted. Consequently, certain studies that did not fall within the scope of business and finance were identified and subsequently excluded from further analysis. From the initial pool of 663 studies, 156 duplicate and irrelevant articles were excluded, and 25 studies were removed due to the unavailability of full-text access. As a result, 482 studies proceeded to the eligibility stage, where further assessment was conducted to determine their suitability and relevance for the systematic review.

2.3. Eligibility

During this stage, a meticulous examination of the full text of the selected studies was carried out to determine their eligibility for inclusion in the review. This phase involved further filtering of the previously identified studies by applying specific inclusion and exclusion criteria.

Inclusion Criteria:

- Studies that utilize machine learning or intelligent techniques for the purpose of predicting a potential bankruptcy.
- Studies that include a comparison of different machine learning methods with statistical methods for bankruptcy prediction.

Exclusion Criteria:

- Studies that solely rely on the Altman method (1968) exclusively employ statistical methods for predicting a potential bankruptcy.
- Studies that lack sufficient details regarding the machine learning models employed in their analysis. In particular, there were no clear details regarding the tuning of the parameters of each model or the methods used for feature selection.
- Studies that did not utilize financial distress data as part of their research. In particular, we excluded all studies that did not employ financial distress data because of their low predictability power in bankruptcy models.
- Studies that were presented in conferences.

After a careful examination of each remaining study, 222 relevant studies were finally deemed eligible for inclusion in the systematic review. These studies met the established criteria and were considered in the final stage.

2.4. Inclusion

The final stage of this study involves the creation of a database for conducting a quantitative analysis. However, 15 out of the 222 studies were classified as reviews. While these review studies have been considered for the purpose of the current study, due to their nature, they were excluded from the quantitative analysis. Consequently, the remaining 207 studies were acknowledged as empirical studies and were used for the quantitative analysis. All data analysis was conducted using the Microsoft Excel.

For each of the 207 empirical studies, comprehensive details were gathered, including the study’s title, the names and number of authors, the year of publication, the name of the publishing journal, author-specified keywords, the machine learning models employed in the research, the performance metrics used to assess the models, the validation methods applied, and other useful information. The flow diagram in Fig. 1 illustrates the step-by-step process that was adopted in the study, following the PRISMA guidelines.

Adhering to the PRISMA guidelines and carefully selecting empirical studies for the quantitative analysis, this study aims to provide valuable insights into the application of machine learning and intelligent techniques in predicting business failure. The systematic approach employed in this study enhances the reliability and validity of the results and facilitates a comprehensive and unbiased review of the available literature in this domain.

3. Review of the selected literature

3.1. Overview of the methods in bankruptcy prediction

Bankruptcy prediction can be considered a classification problem that involves a two-phase process. In the first phase, a model is trained using a dataset consisting of tuples and attributes. One of the attributes, known as the class-label attribute, indicates the predefined class to which each tuple belongs, such as bankrupt or non-bankrupt. This training stage is referred to as supervised learning. In the second phase, the model is used to classify new instances, which are not part of the training dataset but belong to the validation set. If the model demonstrates an acceptable level of accuracy on the validation set, it is considered effective at distinguishing new, real-world cases.

In the field of bankruptcy prediction, various techniques have been employed to forecast future bankruptcies, such as decision trees (DTs), support vector machines (SVMs), neural networks (NNs), k-nearest neighbor (kNN), Bayesian Networks (BN), and genetic algorithms (GA). Fig. 2 illustrates the most frequently used machine learning techniques in the selected papers.[3] It is observed that the most frequently selected method is Neural Networks. However, the current research trend in this field revolves around the development of hybrid or ensemble classifiers, which aim to improve the precision of predictions. Ensemble classifiers combine multiple classifiers to enhance accuracy, surpassing the performance of individual classifiers that may be less accurate (Tsai, 2014). Many researchers have developed ensemble models for this scope (Muslim & Dasril, 2021; du Jardin, 2021b; García et al., 2019; Liang et al., 2018; du Jardin, 2016; Wang et al., 2014; Yu et al., 2014). Hybrid classifiers integrate two or more machine learning techniques, particularly clustering and classification techniques, in a sequential manner (Tsai, 2014). By integrating multiple data sources and modeling techniques, hybrid models have the potential to provide more comprehensive and reliable predictions of business failure (Faris et al., 2020). Table 1 provides indicative research studies that have developed hybrid and ensemble models.Table 2

[3] The classification of studies into hybrid or ensemble classifiers was performed by the authors based on the information provided in the selected studies.

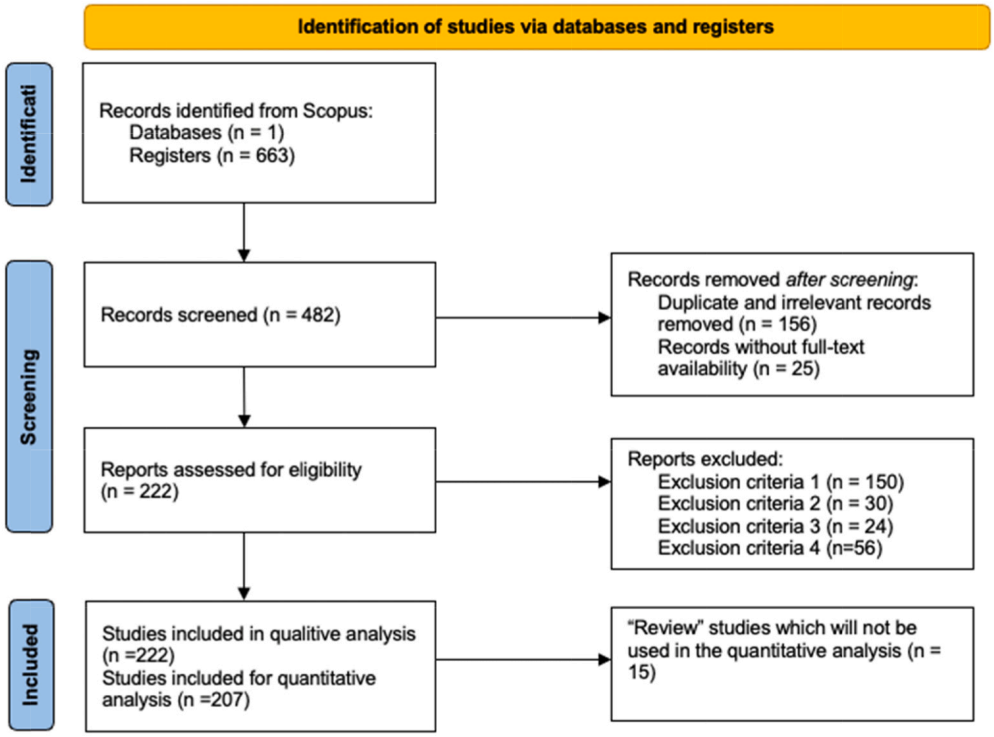

Fig. 1. PRISMA flow diagram. http://www.prisma-statement.org/PRISMAStatement/FlowDiagram?AspxAutoDetectCookieSupport=1.

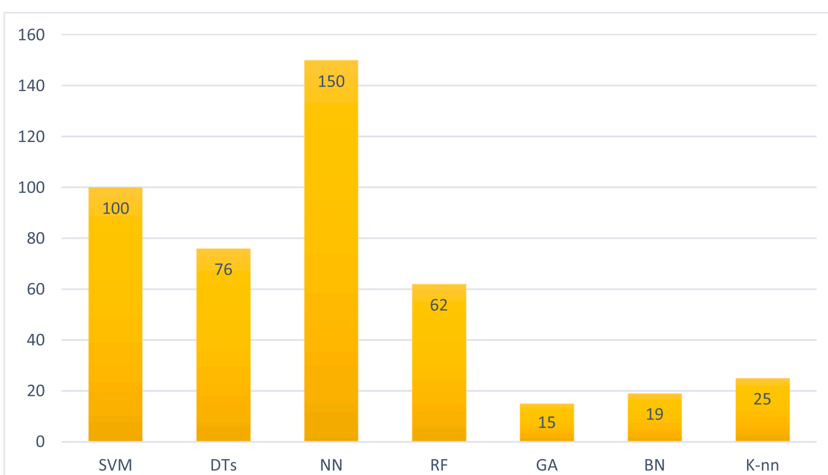

Fig. 2. Most frequent ML methods. .
Source: Author's compilation

After careful examination of all the studies, it has been observed that the most effective ensemble methods are Boosting and Bagging methods. Out of the 48 studies that developed ensemble models, Extreme Gradient Boosting (XGBoost) was the most frequently used method (25 studies), followed by AdaBoost, Gradient Boosting, Bagging, CART, CATBoost, LightBoost and CHAID. For example, Qian et al. (2022) proposed an ensemble model that integrates a heuristic method called Permutation Importance (PIMP) with the XGBoost algorithm. The authors utilized a dataset of Chinese listed companies in financial distress, containing financial variables from 2001 to 2019. The proposed model achieved an average AUC of 89 %, representing an improvement of 6.42 % compared to single classifiers.

Business failure is a complex phenomenon influenced by multiple factors, including financial indicators, market trends, and macroeconomic variables. Hybrid models possess the advantage of capturing the multidimensionality of these factors and their interactions, enabling a more holistic understanding of the drivers of business failure. Through robust predictions, businesses, investors, and policymakers can proactively identify companies or sectors at high risk of failure, enabling them to take preemptive actions to mitigate risks and make well-informed strategic decisions (Faris et al., 2020; Chung et al., 2016). Therefore, this section discusses the hybrid models developed in the selected papers, showcasing their potential in bankruptcy prediction. Chen (2012b) developed a hybrid model that combines Support Vector Machines (SVM) and Self-organizing Maps (SOM) to forecast a company's future trends. When compared to single models (SOM, SVM, and LVQ algorithm), the suggested model performed better.

Chang et al. (2013) proposed a hybrid model that combines Kernel

Table 1
Hybrid & Ensemble Models.

Type of model	Definition	Number of articles / %	Example	Context
Hybird	Hybrid models integrate two or more machine learning techniques, particularly clustering and classification techniques, in a sequential manner (Tsai, 2014).	55/ 27 %	Chen (2014)	Chen (2014) integrated the PSO method with SVM to develop a hybrid model capable to provide high predicted accuracy in the field of bankruptcy prediction. The proposed model achieved accuracy 95 %, precision 94.73 % and specificity 95.23 %.
Ensemble	Ensemble models combine multiple classifiers to enhance accuracy, surpassing the performance of individual classifiers that may be less accurate (Tsai, 2014)	48 / 20 %	Song & Peng (2019)	Song & Peng (2019) developed ensemble models based on SMOTE method in order to predict corporate bankruptcy. The ensemble methods employed in their research included boosting, decision trees and neural networks. The SMOTE based ensemble method which have been outperformes was the the SMOTEBoost-C4.5 which achieved accuracy 96.19 %.

Source: Author's compilation

local Fisher discriminant analysis (KLFDA) and a manifold-regularized SVM (MR-SVM). KLFDA was used to find an optimal projection that maximizes the distance between data points from different classes at each local area of the data manifold, whereas MR-SVM was used to warp the structure of feature space data dependently in order to reflect the underlying geometry of the data manifold.

Li & Sun (2012) examined the problem of imbalanced data using a dataset of publicly listed Chinese hotel companies and developed a model that combines the nearest neighbor with support vector machines to provide more accurate predictions. Shie et al. (2012) suggested a hybrid model that combines Particle Swarm Optimization and Support Vector Machines. Particle Swarm Optimization is widely used in problem-solving where the best solution can be represented as a point or in a d-dimensional space. In addition, the authors proposed an adaptive inertia weight mechanism to increase the performance of the PSO algorithm.

Chuang (2013) developed a Case-based reasoning (CBR) hybrid model in order to improve business failure predictions. The CBR has been combined with other classification methods. The proposed models are the following: RST–CBR (Rough Set Theory with CBR, RST–GRA–CBR (integrating RST, Grey Relational Analysis, and CBR), and CART–CBR (combining Classification and Regression Tree with CBR). Li & Sun (2012) also used the CBR method and developed ensemble models. The aim of Li & Sun (2012) was not to create new models but to compare the CBR approach to existing ensemble methods.

Lin et al. (2013) developed a hybrid model that predicts firm failure by combining the Locally Linear Embedding (LLE) technique and Support Vector Machines (SVM). LLE is an unsupervised method that can be used for dimension reduction. Tsai & Hsu (2013) proposed a stacked model or a *meta*-learning model. The model has two levels. The first level, 0, filters out the noise of the training data. In this way, data are represented in a better way and at the second level, 1, the classifier uses this data to achieve better performance. The authors used two levels 0 and three-level 0 classifiers. The three-level 0 classifiers are based on MLP, CART and LR and the two-level 0 classifiers are based on the top two single baseline classifiers among the three classifiers to provide a better prediction accuracy rate. The same procedure follows the level 1 classifier.

Zhang et al. (2013) suggested a hybrid model to predict company bankruptcy. The model has three steps: Feature extraction, rule-based modeling, and the genetic ant colony algorithm (GACA). To identify the ideal parameters for the rule-based model, the authors proposed a new form of algorithm that employs the fitness scaling approach and the chaotic operator.

Chen (2014) integrated the PSO method with SVM to develop a hybrid model capable to provide high predicted accuracy in the field of bankruptcy prediction. The author compared his proposed model with the Grid-SVM model. Indeed, the proposed model achieved better accuracy.

Tsai (2014) developed a novel hybrid model, combining clustering techniques with ensembles classifiers. The clustering techniques are the Self Organizing Maps (SOM) and the k-means. The single classifiers are Logistic Regression, MLP neural network, and Decision Trees. The ensemble method that was used is the weighted voting. In total, Tsai (2014) formed 21 different hybrid models. The experimental results showed that combining SOM with classifier ensembles such as the weighted voting method can improve the prediction result and low the Type I & II errors. Tsai et al. (2014) trained different models with ensemble techniques, boosting and bagging. In this way, they constructed ensemble classifiers. The selected techniques were Neural Netwrok, SVM and DTs.

Zhou et al. (2014) suggested a new approach for the optimization of support vector machines. Moreover, with the new approach they tried to optimize the feature selection procedure. Zhou et al. (2014) developed two hybrid models: GA-based features selection and parameter

Table 2
FInancial variables used in bankruptcy prediction models.

Liquidity = working capital/ total assets
Profitability = retained earnings/ total assets
Productivity = EBIT/ total assets
Debt ratio = market value of equity/ total debt
Asset turnover = sales/assets
Operational margin = EBIT/sales
Assets growth = total assets$_t$ – total assets$_{t-1}$ / total assets$_{t-1}$
Sales growth = total sales$_t$ – total sales$_{t-1}$ / total sales$_{t-1}$
Growth of employees = number of employees$_t$ − number of employees$_{t-1}$/ number of employees$_{t-1}$
Change in ROE = ROE$_t$ −ROE$_{t-1}$
Change in price to book = price to book$_t$ – price to book$_{t-1}$

Source: Author's compilation.

optimization for SVM (GA-SVM), and Direct search-based features selection and parameter optimization for SVM (DS-SVM).

To anticipate a potential bankruptcy, Kim et al. (2016) developed hybrid models combing clustering methods, genetic algorithms (GA), and neural networks (NN). The goal of using clustering is to create the optimal training set, and the combination of GA and NN is to achieve a balance between the minority and majority classes.

Azadnia et al. (2017) suggested a hybrid model that included fuzzy theory and NN. The model is highly precise and can assist investors in making decisions. Chou et al. (2017) created a hybrid model that combined fuzzy theory with genetic algorithms. Ekinci & Erdal (2017) developed ensemble and hybrid models. The base classifiers were Logistic, J48 and Voted Perceptron. The ensemble methods were Random Subspaces, Bagging and Multiboost. The authors aimed at improving the prediction accuracy for bank bankruptcy.

Fallahpour et al. (2017) developed an ensemble model using SVM and the Sequential Floating Forward Feature Selection (SFFS) method aiming to create an algorithm that would achieve high accuracy in predicting financially distressed companies. In this method, an ideal subset is generated by adding a feature to the previously selected subset of features or eliminating a particular feature. This method is capable to reduce the probability to stuck in a local optimum, resulting in higher accuracy compared to other feature selection methods. The proposed model, SFFS-SVM, could increase the accuracy of the prediction by more than 11 % compared with a single SVM classifier. In addition, the proposed model could have been useful for feature selection.

Wang & Wu (2017) developed an ensemble model called KFSOM, which combines the manifold technique and a fuzzy self-organizing map based on kernels. The main idea behind this model is to address the challenges of high-dimensional data by reducing the dimensionality through the manifold technique and using KFSOM as an optimization algorithm to improve the model's performance.

Using Kohonen maps, du Jardin (2018) proposed ensemble bankruptcy models. A Kohonen map is a self-organizing map in which each neuron is represented by a set of weights, with each weight corresponding to a variable that characterizes a sample of observations. The author utilized the following ensemble methods: bagging, boosting, random subspace and rotation forest.

Affes & Hentati-Kaffel (2019) developed a hybrid model that combines K-means and the Multivariate Adaptive Regression Splines (MARS) model in order to reduce the misclassification problem in bankruptcy prediction. The main advantage of this model is its capacity to explore the complex nonlinear relationships between response variables and various predictor variables.

Le et al. (2019a) proposed the bankruptcy prediction model, gXGBS, which utilizes the squared logistics loss (SqLL) for GPU-based extreme gradient boosting machine (XGBS). The model aims to improve bankruptcy predictive performance. In the first step, the XGBS algorithm incorporates the custom loss function, SqLL, to enhance the boosting process. In the second step, gXGBS employs GPU computing for accelerating decision tree construction. Finally, for large datasets, the third step applies the histogram-based tree construction algorithm using multi-GPU, named gXGBS_hist, as an approximation algorithm.

Le et al. (2019b) suggested a hybrid model for bankruptcy prediction that integrates the oversampling technique and a cost-sensitive learning framework. The suggested model adjusts the class distribution of the Korean dataset with a certain balancing ratio using the SMOTE-ENN (Synthetic Minority Oversampling technique – Edited nearest neighbor) and then predicts bankruptcy using the CBoost algorithm (Cluster Based Boosting method).

Ansari et al. (2020) developed MOA-PSO, a hybrid approach based on Magnetic Optimization Algorithm (MOA) and Particle Swarm Optimization (PSO) for training Neural Networks in bankruptcy prediction. MOA-PSO combines MOA's local search capabilities with PSO's social thinking power. De Bock et al. (2020) proposed a methodological framework for generating heterogeneous ensemble classifiers via ensemble selection as a cost-sensitive strategy for building business failure prediction models.

Tsai (2020) predicted bankruptcy using a two-stage hybrid learning architecture. The first stage is data pre-processing, in which a clustering method is employed to find representative training instances from each class. The clustering results are then utilized to build a classifier or a prediction model in the second stage. The authors employed the k-means and AP (Affinity Propagation) algorithms for clustering and the SVM and LR as basis classifiers. In terms of AUC, SVM performs the best.

Uthayakumar et al. (2020) proposed a cluster-based classification model that consists of two stages: improved K-means clustering and a classification model based on the fitness-scaling chaotic genetic ant colony algorithm (FSCGACA). The K-means method eliminates incorrectly clustered data in the first stage, then a rule-based model is fitted to the given data, and finally, the FSCGACA identifies the optimal parameters for the rule-based model.

Aljawazneh et al. (2021) examined the performance of three well-known Deep Learning methods (Long-Short Term Memory, Deep Belief Network and Multilayer Perceptron model of 6 layers) with three bagging ensemble classifiers (Random Forest, Support Vector Machine and K-Nearest Neighbor) and two boosting ensemble classifiers (Adaptive Boosting and Extreme Gradient Boosting) in corporate financial failure prediction. They also used extremely unbalanced data. To deal with the imbalanced problem, they also used five oversampling techniques, two hybrid balancing techniques, and one clustering-based balancing technique. The results showed that the Multilayer Perceptron model of 6 layers, in conjunction with SMOTE-ENN balancing method, achieved the best performance in terms of accuracy, recall and type II error metrics.

He & Fan (2021) developed a hybrid ensemble model to improve the default prediction performance. First, a tree-based approach (LightGBM) is utilized to learn new feature interactions and improve the representation of original features. Second, to develop deeper feature interactions, a deep learning method (Inner Product-based Neural Network) is applied as a feature-generating method. Third, to achieve higher predictive outcomes, an ensemble learning method is employed to integrate the deep learning classifier with tree-based classifiers.

Kim & Upneja (2021) employed the recurrent neural network (RNN) and long short-term memory (LSTM) to improve corporate bankruptcy predictions using sequential data. Based on the metric AUC, the RNN and LSTM methods have better forecasting performances relative to the logistic regression, support vector machine, and random forest methods. However, the RNN and LSTM methods cannot indicate the importance of each explanatory variable.

Safi et al. (2022) proposed a cost-sensitive model to improve the prediction of minor classes in a financial distress dataset and then apply majority voting ensemble learning to create a strong learner out of several weak learners. The proposed model was based on a Particle Swarm Optimizer (PSO) and a Competitive Swarm Optimizer (CSO) with a cost-sensitivity fitness function.

Cheng & Hoang (2015) developed a hybrid model, FICDP, for Contractor Default Prediction. It combines Fuzzy K-nearest Neighbor Classifier (FKNC), Synthetic Minority Over-sampling Technique (SMOTE), and Firefly Algorithm (FA). The SMOTE addresses class imbalance, FKNC assigns fuzzy memberships based on distances, and FA optimizes model parameters. FICDP enhances prediction performance for contractor default in construction, tackling industry complexities and economic fluctuations.

3.2. Input data and imbalanced data

The selection of the data sample and the appropriate variables is critical for the development and training of a model and allows us to draw conclusions from various scenarios useful for decision-making. We observe that 21 out of the 207 papers used bank data such as the following articles (Affes & Hentati- Appiahene et al., 2020; Manthoulis et al., 2020; Shrivastav & Janaki Ramudu, 2020; Viswanathan et al., 2020; Kaffel, 2019; Halteh et al., 2018; Ecer, 2013; Garcia-Almanza et al., 2013;). For 7 studies we don't know the type of the sector and the remaining publications used data from other industries, such as manufacturing, construction e.g. (Ling & Cai, 2022; Jang et al., 2021; Gregova et al., 2020;; Vochozka et al., 2020; Karminsky & Burekhin, 2019; Alaka, Oyedele, Owolabi, Bilal, et al., 2018; Andreano et al., 2018; Tian & Yue, 2017; Slavici et al., 2016; Heo & Yang, 2014), agriculture companies (Klepac & Hampel, 2017), and restaurant industry (Becerra-Vicario et al., 2020).

However, the sector specificity is not consistently expressed in all studies, with some merely indicating that the sample comprises firms without specifying the industry. Fig. 3 illustrates the distribution for firms and banks, revealing that the banking sector (21 cases) is under-researched compared to the non-banking sector (181 cases). Fig. 4 provides insights into the geographic distribution of data samples. It is evident that Poland has been extensively examined, which is largely attributed to publicly available datasets in Poland. Apart from Poland, data from Germany, Taiwan, Japan, and France are also publicly available. In summary, the analysis reveals three main clusters: studies employing European Union (EU) data, studies centered on the USA, and studies utilizing data from the Middle East. This observation underscores the need to delve into the distinct cultural and contextual factors contributing to bankruptcy across different regions. Moreover, it underlines the challenge of data accessibility, particularly within the banking sector, emphasizing the need for further research to address these data gaps and explore the various factors influencing financial outcomes across different cultural contexts.

The dataset size is crucial for bankruptcy prediction because of two main reasons. First, a larger dataset provides more information and observations, enabling a more comprehensive analysis of bankruptcy patterns and trends. This leads to more accurate predictions as models capture a wider range of scenarios. Second, a larger dataset helps address the issue of data imbalance, commonly encountered in bankruptcy prediction where bankrupt companies are fewer. A small dataset worsens this imbalance, hindering the model's ability to learn and generalize effectively. Fig. 5 shows that 127 studies employed a sample size of up to 1,000 observations. 47 studies utilized a data sample in the range of 15,000 to 100,000 observations, while only 17 studies utilized a sample size exceeding 100,000 observations. This suggests an under-utilization of larger datasets in the examined studies. The lack of extensive datasets may constrain the potential robustness of predictions. In contrast, the use of larger datasets is alleged to enhance the reliability and generalizability of predictive models, thus warranting further investigation.

Several studies have utilized multiple datasets to validate their bankruptcy predictions (Aly, Alfonse, Roushdy, et al., 2022; Uthayakumar, Metawa, et al., 2020; Lin et al., 2019; Abellán & Mantas, 2014)l., 2020; Lin et al., 2019; Abellán & Mantas, 2014) incorporated multiple datasets in their analyses. Lin et al. (2019) employed three datasets, namely the Polish, German, and Japanese datasets. Jang et al. (2019) expanded their analysis to four datasets, including the Polish, German, Australian, and Japanese datasets. Uthayakumar et al. (2020) utilized five datasets, including the Qualitative, AnalcatData, Australian, German, and Polish datasets. Durica et al. (2019) used the Polish dataset, which has been used also from multiple authors. The authors utilized multiple datasets to ensure the robustness and generalizability of their bankruptcy prediction models (Elhoseny et al., 2022; Antunes et al., 2017). Notably, 21 out of 40 studies employing multiple datasets incorporated at least one publicly available dataset, such as Qualitative,[4] Analcat,[5] Australian,[6] German,[7] and/or Polish.[8] Specifically, the Analcat dataset comprises 50 cases, equally divided into bankrupted and non-bankrupted. The Qualitative dataset consists of 107 cases recorded as bankrupted and 143 as non-bankrupted. The Australian dataset comprises 383 cases recorded as bankrupted and 307 as non-bankrupted. The Polish dataset includes 2,091 cases recorded as bankrupted and 41,314 as non-bankrupted. Finally, the German dataset comprises 300 cases recorded as bankrupted and 700 as non-bankrupted.

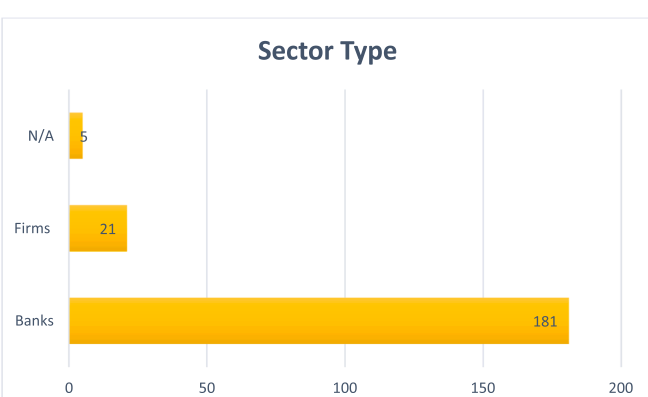

Fig. 3. Distribution of sector type. .
Source: Author's compilation

200 out of 207 papers employed financial ratios. Table 2 summarizes the financial variables that have been utilized in almost all studies, along with other variables to predict bankruptcy. The first five variables presented in Table 2 have been derived from the Altman method and have consistently demonstrated their significance. Prior researchers have extensively used Altman variables due to their high predictive power and applicability across industries. However, some studies incorporated macroeconomic factors, such as those conducted by Chen et al. (2023), Schalck & Yankol-Schalck (2021), Jencova et al. (2021), Lahmiri et al. (2020), Le et al. (2019a), Liang et al. (2016), and Chen (2012a, 2012b). In specific, Chen et al. (2023) added variables taken from annual reports and achieved to improve both the effectiveness of the developed bankruptcy prediction and the F1 metric. Non-financial ratios, including market ratios, management competence and ability, government policies and human resource management, were also employed. Table 3 summarizes the non-financial/qualitative variables used in bankruptcy prediction models. Lahmiri et al. (2020) demonstrated the effectiveness of utilizing qualitative information in improving the efficiency of artificial neural networks for bankruptcy prediction. Faris et al. (2020) and Sisodia & Verma (2018) considered non-financial variables, such as the number of employees, auditor's opinion and expenses related to court incidents. Faris et al. (2020) argued that adding those extra variables did not increase the accuracy of the developed models, but it gave an insight on the most influencing factors in identifying the bankruptcy cases. The five relevant features were: Number of judicial incidences in total, judicial expense in total, return of equity, asset turnover and working capital. Muñoz-Izquierdo et al. (2019) examined the explanatory power of external audit reports and their model achieved an average accuracy 75 %. Mai et al. (2019) used textual disclosures by using deep learning. Although the developed models achieved accuracy between 71 %-78 %, it was difficult to interpret the results. However, the authors identified many words which related to bankruptcy such as "repurchase", "restated", "indebtedness", "dividend", "tranche", "international", "exit", "compensation" "wages" and "suppliers". Lee et al. (2020) suggested

[4] Qualitative_Bankruptcy (2014). UCI Machine Learning Repository. https://doi.org/10.24432/C52889.

[5] Simonoff, J S. (2003). Analyzing Categorical Data. Springer. https://axon.cs.byu.edu/data/statlib/nominal/analcatdata_bankruptcy.arff.

[6] Quinlan, R. Statlog (Australian Credit Approval). UCI Machine Learning Repository. https://doi.org/10.24432/C59012.

[7] Hofmann, H. (1994). Statlog (German Credit Data). UCI Machine Learning Repository. https://doi.org/10.24432/C5NC77.

[8] Tomczak, S. (2016). Polish Companies Bankruptcy. UCI Machine Learning Repository. https://doi.org/10.24432/C5F600.

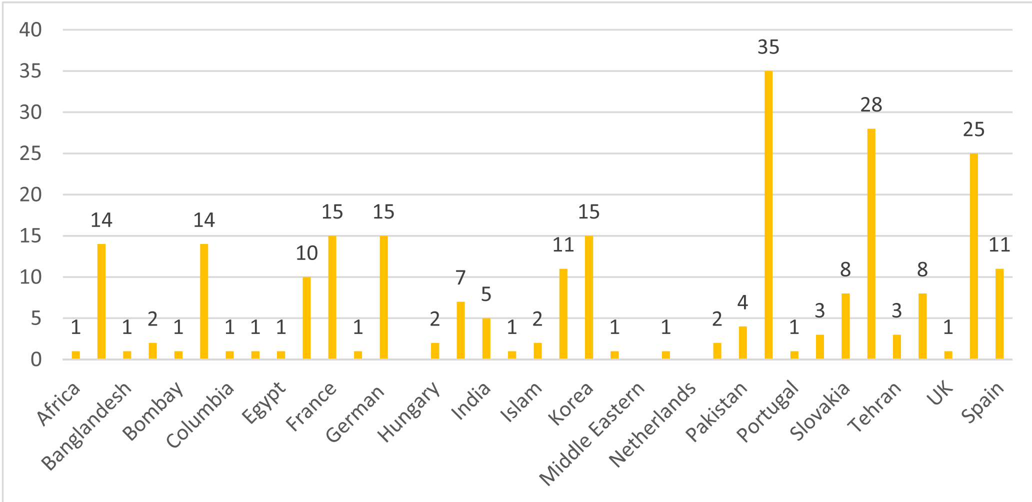

Fig. 4. Geographic Distribution of data samples. .
Source: Author's compilation

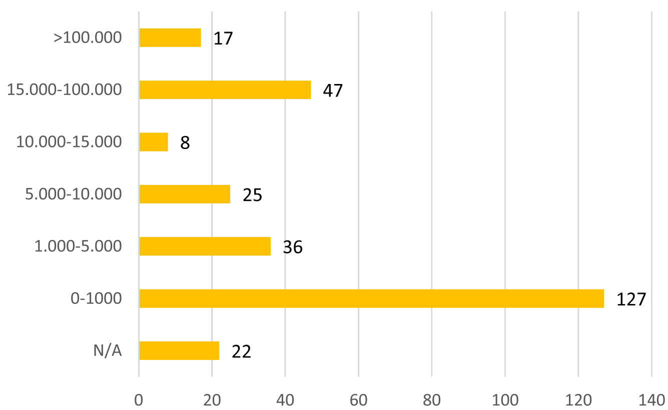

Fig. 5. Data sample size distribution. .
Source: Author's compilation

those variables such as financing ability, CEO's dependability and future profitability for bankruptcy prediction within a time horizon of less than three years, while variables such as credit status, financing ability and competitive strength were deemed important for bankruptcy prediction beyond three years. By adding those variables, the accuracy increased by 2 %. Furthermore, only two studies incorporated non-financial variables such as textual information. In specific, De Jesus & Besarria (2023) and Zhang et al. (2022) integrated textual data. Zhang et al. (2022) focused on the emotional tone of financial news texts, assessing its impact on prediction accuracy. 13 models were developed using both financial data and textual information. The researchers observed an average prediction accuracy increase of 0.0331 when incorporating the emotional tone of financial news compared to models utilizing solely financial data. In addition to accounting data, De Jesus & Besarria (2023) included macroeconomic variables to predict potential bank insolvency. Their approach incorporated sentiment indices derived from quarterly reports of bank institution managers. The authors identified key terms, such as "securities," "expenses," "covid," and "payments," with substantial effects on bank risk.

In addition, several papers e.g., (Bateni & Asghari, 2020; Faris et al., 2020; Lee et al., 2020; Sisodia & Verma, 2018) highlighted the potential benefits of incorporating non-financial ratios in bankruptcy prediction models to enhance predictions and aid decision-making for managers. Principal Component Analysis (PCA), ANOVA tests, t-statistics, and Factor Analysis were some of the techniques employed for feature selection using non-financial ratios (Dube et al., 2023; Venkateswarlu et al., 2022; Farooq & Qamar, 2019; Nguyen et al., 2019; Erdal & Ekinci, 2013; Erdogan, 2013; Chen, 2013). In total, 100 out of 207 studies employed one of the aforementioned methods providing an explanation for the importance of variables used. While the number of studies attempting to clarify variable significance is substantial, there is still an uncertainty about the results. This uncertainty is intertwined by the prevalent use of artificial intelligence (AI) methods, which often function as black boxes. As highlighted in the introduction, the opacity of AI models raises concerns as their internal mechanisms are complex and challenging to interpret.

However, during the examination of the selected data that the researchers chose, the issue of imbalanced data became apparent. Our

Table 3
Non-financial/qualitative variables used in bankruptcy prediction models.

Number of employees
Age of the company
Delay: if the company has submitted its annual accounts on time
Historic number of judicial incidences and amount of money spent on judicial incidences
Industry risk: Government policies and internation agreements, cyclicality, degree of competition, the sensitivity to changes in macroeconomic factors, the size and growth of market demand
Management ability and relationships: Ability and competence of management, stability of management, the relationship between management/owner, human resources management, growth process/business performance, achievement and feasibility, CEO's reliability, CEO's professionalism
Finnacing options: Direct financing, indirect financing, other financing
Competitiviness: Market position, level of core capacities, differentiated strategy
Number of partners
Technical ability: R&D investment, possibility of technology expansion, core technology superiority/discrimination

Source: Author's compilation.

systematic literature review process reveals that many authors attempted to overcome the problem of unbalanced data by utilizing oversampling or undersampling methods. When confronted with the problem of business failure, bankrupt cases are fewer compared to non-bankrupt cases. Imbalanced data have asymmetry between the two classes. Specifically, the non-bankrupt cases constitute the majority class, with a significantly larger number than the bankrupt firms, which represent the minority class (Zou et al., 2022).

It is essential for researchers to examine the issue of unbalanced data in bankruptcy prediction for several reasons. First, predictive models trained on unbalanced data tend to be biased towards the majority class, compromising the model's ability to accurately predict bankruptcy cases. This bias is known as performance bias and can lead to inaccurate results (Safi et al., 2022). We cannot posit clearly whch proportion between the two classes are problematic, however, after a thorough examination of the selected studies we noticed that the diffrenece between the two classes are extremely high. For instance, Song and Peng (2019) and Jang et al. (2019) used the Polish dataset which contains 41,314 non-bankrupted cases and 2,091 bankrupted cases. It is worth mentioning that the Polish dataset has been extensively used by prior studies due to the noticeable ratio of non-bankrupted and bankrupted cases. Second, misclassification costs are not equal for both classes in bankruptcy prediction. Misclassifying a bankrupt company as non-bankrupt may result in severe financial consequences, while misclassifying a non-bankrupt company as bankrupt may lead to inconvenience, but not necessarily to significant financial losses. This cost imbalance underscores the importance of addressing unbalanced data in bankruptcy prediction (Y. Liu et al., 2022). Third, bankruptcy prediction models are often used to support decision-making processes, such as assessing credit risk or making investment decisions. If the model is not effectively trained to handle unbalanced data, it may provide inaccurate predictions, leading to flawed decisions. Addressing the issue of unbalanced data is pivotal to developing bankruptcy prediction models that are more reliable, accurate, and useful in real-world applications, ultimately assisting stakeholders in making informed decisions and managing financial risks effectively (Papíková & Papík, 2022). Therefore, understanding the impact of unbalanced data and addressing it appropriately can improve the reliability and usefulness of these models.

In total, 38 studies have employed various strategies to address the issue of imbalanced data in bankruptcy prediction. Liu et al. (2022) compared three sample methods using seven machine learning approaches: Oversampling, Undersampling and Synthetic Minority Oversampling Technique (SMOTE). SMOTE is a data oversampling technique that generates samples from the minority class to achieve a more balanced or nearly balanced training dataset (Liu et al. (2022). Among these methods, SMOTE performed the best, demonstrating its ability to forecast financial distress based on unbalanced data. Tumpach et al. (2020) also employed the SMOTE technique to handle significantly unbalanced datasets from Slovak enterprises. Song & Peng (2019) employed random undersampling (RUS) and SMOTE techniques. RUS removes the majority examples from the original data to reduce the imbalance.

Since 2018, an increasing number of studies have investigated the use of sampling approaches to provide bankruptcy predictions. The authors of these studies have argued for additional research in this area. Table 4 illustrates indicative studies that have examined the aforementioned methods.

Table 4
Sampling techniques.

Type of method	Definition	Number of articles / %	Example	Context
Oversampling	The oversampling technique is designed to mitigate class imbalance by generating synthetic data points within the minority class. The objective is to establish a more equitable balance between the minority and majority classes (Adisa et al. 2023)	24 / 9 %	Adisa et al. (2023)	Adisa et al. (2023) introduced a novel model that integrates Particle Swarm Optimization (PSO) with Long Short-Term Memory (LSTM) neural networks for the purpose of predicting bankruptcy. To address the issue of imbalanced data, they applied the Synthetic Minority Over-sampling Technique (SMOTE). The proposed model exhibited a noteworthy accuracy of 90.51 % when SMOTE was applied with a 10 % oversampling rate.
Undersampling	The primary concept behind under-sampling techniques is to address class imbalance by re-sampling the majority class. In this approach, diverse data points from the majority class are deliberately removed, aiming to achieve a more balanced class distribution by reducing the number of samples in the majority class to match that of the minority class (Adisa et al. 2023)	26 / 10 %	Wang & Liu (2021)	Wang and Liu (2022) addressed the problem of imbalanced data using several methods: Tomek Links, Edited Nearest Neighbors, Repeated Edited Nearest Neighbors, One-Sided Selection and the Neighbor Cleaning Rule. They evaluated the performance of these methods using the F2-measure. The model combined Naive Bayes (NB) with Edited Nearest Neighbors and achieved the highest F2-measure value (0.423).

Source: Author's compilation.

3.3. Metrics

The selection of an appropriate performance metric is a crucial aspect when evaluating a proposed model. Researchers must consider various factors before deciding on the metric to be used. First, the nature of the problem being addressed should be taken into account. In the case of bankruptcy prediction, as already mentioned, the number of bankrupt banks or firms is generally lower than the number of non-bankrupt banks or firms. This number imbalance can influence the choice of an appropriate metric.

Another important aspect is the risk of data overfitting. Overfitting occurs when a model performs exceptionally well on the training data but fails to generalize to new, unseen data. To mitigate this risk, researchers need to select a performance metric that can effectively assess the model's generalization ability. Several performance metrics address these concerns. Accuracy, Type I error (false positive rate), Type II error (false negative rate), sensitivity, specificity, and the area under the curve (AUC) are widely employed in evaluating classification models. These metrics provide insights into different aspects of model performance, such as overall accuracy, ability to correctly identify positive cases (bankrupt banks or firms), and ability to correctly identify negative cases (non-bankrupt banks or firms). The AUC metric, represented by the receiver operating characteristic (ROC) curve, offers a comprehensive evaluation of the model's performance across various classification thresholds.

Another method to address overfitting is to split the dataset into training, validation, and test sets, along with the utilization of n-fold cross-validation. In n-fold cross-validation, the data is divided into n equal-sized subsamples, where subsets are used as training sets and the remaining subsets serve as the testing set. This approach helps to reduce bias and variance since a significant portion of the data is utilized for both learning and testing purposes (Lahmiri et al., 2020; Lin et al., 2012).

In addition to cross-validation, other studies (Lombardo et al., 2022; Erdogan et al., 2021) adopted a different strategy by training the model from a specific horizon and using a distinct period as the validation set. This approach allows researchers to assess the model's performance on unseen data, providing a more robust evaluation. Furthermore, 50 studies divided the initial data sample into three parts: training, validation and test sets. This partitioning enables researchers to train the model on the training set, fine-tune its parameters using the validation set, and then evaluate its performance on the independent test set.

Many researchers have attempted to construct models to achieve high accuracy and assess the prediction performance of each model. However, relying solely on accuracy as a metric can provide an overall picture of the model's performance, but lacks specific details, such as the number of correctly predicted cases. This becomes more challenging in the context of imbalanced data, where the proportion of positive and negative cases differs significantly in bankruptcy prediction. Therefore, it is preferable to employ metrics that can offer more informative insights. For instance, metrics such as sensitivity (true positive rate), specificity (true negative rate), Type I error, Type II error and area under the curve (AUC) have been widely used due to the imbalanced nature of bankruptcy prediction. Researchers employed accuracy as a performance metric in all 207 papers, while a subset of researchers adopted additional metrics to ensure more robust results. Fig. 6 illustrates the distribution of sensitivity (true positive rate), specificity (true negative rate), Type I/II error, area under the curve (AUC), and F1 score.

Sensitivity, also known as Recall, measures the ability of a model to correctly predict bankrupt instances. Specificity assesses the model's capability to predict non-bankrupt instances accurately. These metrics

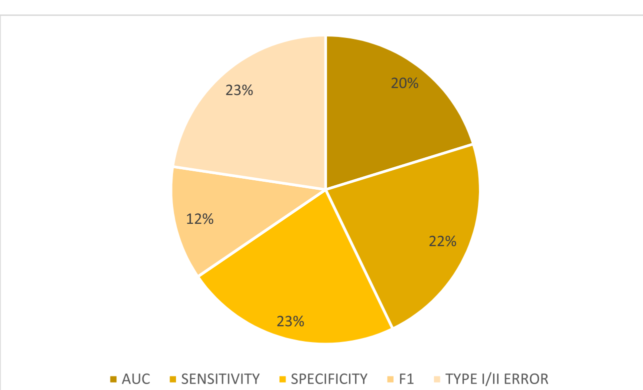

Fig. 6. Distribution of performance metrics. .
Source: Author's compilation

have been employed in 50 studies. Type I and Type II errors, which represent false positive and false negative results, respectively, are often used together. Type II error is particularly important in decision-making scenarios as misclassifying a bankrupt firm as non-bankrupt has a higher risk. These two indicators have been utilized in 30 studies.

Another frequently used metric is the Area Under the Curve (AUC), which measures the model's performance in terms of the trade-off between true positive rate and false positive rate. AUC has been employed in 44 studies. Additionally, the G-mean, which calculates the geometric mean of accuracies for both classes, has been utilized by 20 studies. The G-mean is maximized when there is a balance between the accuracies of both classes. Furthermore, 28 studies examined the F1 metric, which combines precision and recall (sensitivity) using the Harmonic mean (Smith & Alvarez, 2022).

Another notable finding is the limited use of statistical tests to ascertain the significance of outcomes from various developed models. Table 5 reports the studies that applied statistical tests. This final phase of statistical analysis is crucial for validating the reliability of the models' performance. So far, 12 studies have identified the research gap in the current literature. This limited research calls for more comprehensive statistical validation of proposed models, emphasizing the importance of enhancing the robustness of predictions and contributing to the methodological rigor of studies.

Researchers have recognized the limitations of accuracy as a singular metric and have turned their attention to more informative measures such as sensitivity, specificity, Type I and Type II errors, AUC, and F1 score. These metrics provide a more comprehensive understanding of a model's predictive potential in identifying bankrupt cases, thus making informed decisions. Despite this progress, the limited integration of statistical tests raises concerns about the overall quality and reliability of findings in this domain. Therefore, future studies could benefit from conducting more elaborate statistical tests to ensure the significance of their results. Table 6 presents indicative studies that have employed these metrics; however, the identified research gap calls for further investigation into the statistical validation practices within the field.

Finally, in the Appendix, Table 7 provides a comprehensive overview of the 207 selected articles, presenting key details such as authors, firm category, dataset size, time period, and the types of variables considered

Table 5
Studies that applied statistical tests.

Authors	Title of the study
Kim & Kang (2012)	Classifiers selection in ensembles using genetic algorithms for bankruptcy prediction
Li & Sun (2012)	Forecasting business failure: The use of nearest-neighbour support vectors and correcting imbalanced samples – Evidence from the Chinese hotel industry
Chuang (2013)	Application of hybrid case-based reasoning for enhanced performance in bankruptcy prediction
Li & Sun (2013)	Predicting business failure using an RSF-based case-based reasoning ensemble forecasting method
Lin et al. (2013)	A hybrid business failure prediction model using locally linear embedding and support vector machines
Appiahene et al. (2020)	Predicting Bank Operational Efficiency Using Machine Learning Algorithm: Comparative Study of Decision Tree, Random Forest, and Neural Networks
Hu et al. (2021)	Bankruptcy prediction using multivariate grey prediction models
Schalck & Yankol-Schalck (2021)	Predicting French SME failures: new evidence from machine learning techniques
Elhoseny et al. (2022)	Deep Learning-Based Model for Financial Distress Prediction
Li et al. (2022)	Bank efficiency and failure prediction: a nonparametric and dynamic model based on data envelopment analysis
Zou et al. (2022)	Business Failure Prediction Based on a Cost-Sensitive Extreme Gradient Boosting Machine
Ben Jabeur et al. (2023)	Bankruptcy Prediction using the XGBoost Algorithm and Variable Importance Feature Engineering

Source: Author's compilation.

Table 6
Performance metrics.

Metrics	Number of articles / %	Example
AUC Area under the ROC curve is a performance metric commonly used to evaluate the predictive power of a binary classification model (Karas & Reznakova, 2017)Κάντε κλικ ή πατήστε εδώ για να εισαγάγετε κείμενο.	44 / 17 %	Abdullah (2021)
Sensitivity Sensitivity is defined as the ratio of correctly predicted positive instances (True Positives) to the total number of actual positive instances (True Positives + False Negatives) (Paule-Vianez et al., 2020).	50 / 19 %	Aljawazneh et al. (2021)
Specificity Specificity is defined as the ratio of correctly predicted negative instances (True Negatives) to the total number of actual negative instances (True Negatives + False Positives) (Paule-Vianez et al., 2020).	50 /19 %	Venkateswarlu et al. (2022)
F1 The F1 score is a widely employed performance metric in binary classification. Its primary purpose is to strike a balance between precision and recall, consolidating these two metrics into a single numerical assessment of a model's effectiveness. (Smith & Alvarez, 2022)	28 / 10 %	Safi et al. (2022)
Type I/II error Type I error corresponds to classifying something as positive when it is negative (false positive). Type II error corresponds to classifying something as negative when it is positive (false negative). Balancing these errors is often a crucial consideration when designing experiments, conducting hypothesis tests, or building classification models (Bragoli et al., 2022).	50 / 19 %	Bragoli et al. (2022)

Source: Author's compilation

in each study. Examining the details of the selected articles allows researchers to identify potential gaps or areas with limited coverage within the existing literature.

3.4. Bibliometric analysis

Fig. 7 illustrates the evolution of published articles spanning the years 2012 to mid-2023. Since 2017, a considerable increase in the number of publications has occurred, where the volume of published articles has almost doubled. Notably, until mid 2023, 16 papers have been published, underlining the increased academic interest on bankruptcy predictions.

Fig. 8 presents the top 20 articles based on the number citations, while Fig. 9 depicts the distribution of the top 20 articles across journals. The most frequently cited article is that of Barboza et al. (2017). 23 papers have been published in Expert Systems with Applications, followed by 8 articles in the European Journal of Operational Research, 7 articles in Computational Economics, Journal of Forecasting, and Lectures Notes in Computer Science. 6 papers have been published in Annals of Operation Research, IEEE Access, and Sustainability.

4. Discussion

This paper has conducted a comprehensive literature review on bankruptcy prediction, with a specific focus on the application of machine learning methodologies and models. It has offered valuable insights into the field, carrying significant implications for academics, investors, policymakers, and other stakeholders, including creditors, lenders, and business owners.

Through the investigation of various machine learning techniques and predictive models, the study has unveiled several key findings and illuminated future research directions. These findings underscore the critical significance of accurate and robust machine learning-based prediction models in evaluating the financial health and stability of organizations. This research underscores the growing role of advanced computational approaches in enhancing bankruptcy prediction and financial risk assessment.

Our extensive research underlined the current trend toward the development of hybrid models that combine multiple methodologies and variables to improve bankruptcy prediction accuracy. The variety of

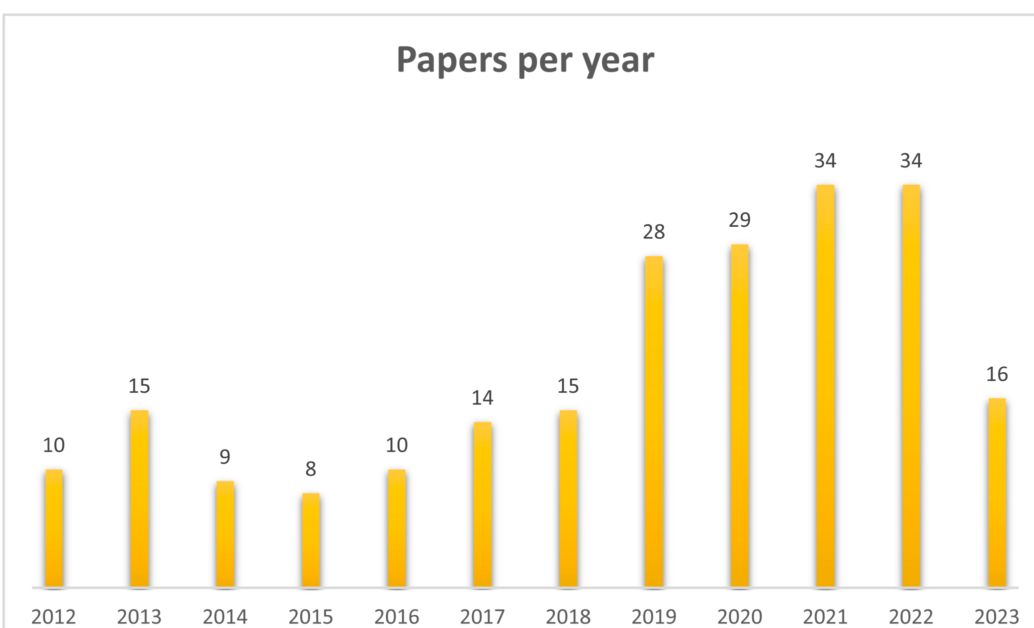

Fig. 7. Published papers per year. Source: Author's compilation

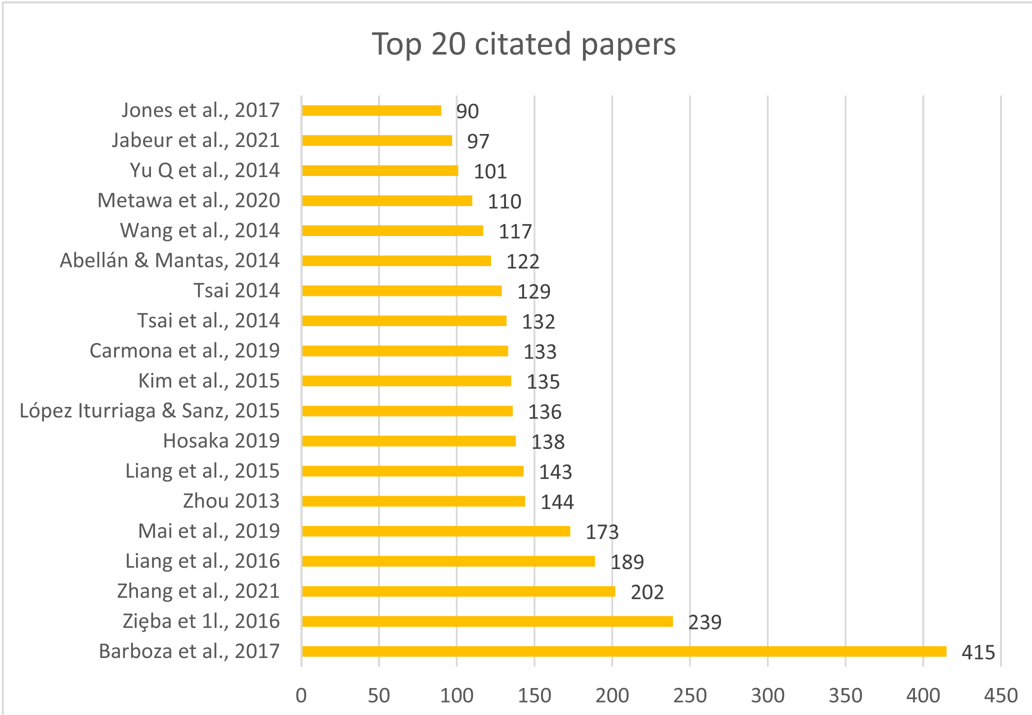

Fig. 8. Top 20 articles based on the number citations. Source: Author's compilation

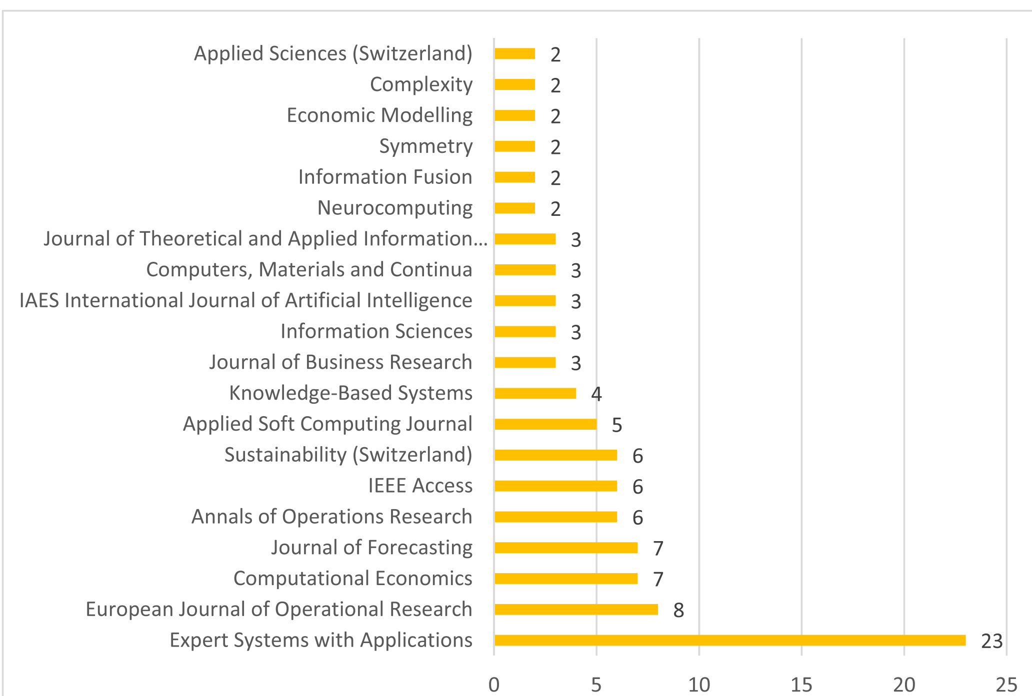

Fig. 9. Distribution of the top 20 articles across journals. Source: Author's compilation

potential combinations remains largely unexplored, presenting an exciting avenue for future research due to the dynamic nature of financial data, which require continuous innovation. In addition, the study of hybrid models will facilitate a deeper investigation into the phenomenon of black boxes. This involves combining the predictive capabilities of advanced machine learning algorithms with the transparency offered by more interpretable models, such as decision trees or rule-based systems. Striking a balance between accuracy and interpretability will be essential to gain the trust of stakeholders, including financial institutions, regulators, and end-users, as the field continues to evolve and contribute to more effective bankruptcy prediction models. Moreover, researchers have attempted to address the problem of imbalanced datasets in bankruptcy prediction by exploring and developing specialized techniques for handling such data. No single methodology has claimed to be superior compared to others. However, the quest for methodological refinement, particularly in the context of unbalanced data, seems like an interesting avenue for future research. Another research gap is the lack of comprehensive statistical tests conducted between the developed models. It is noteworthy that only 12 studies performed statistical tests to assess the significance of their results. This gap calls for a more robust statistical evaluation framework to enhance the reliability and applicability of these models in real world scenarios. Finally, many researchers addressed overfitting using different metrics (i.e., sensitivity, specificity, Type I error, Type II error and Area Under the Curve) and approaches (i.e., splitting the dataset into training, validation, and test sets).

For academics, this review underscores the evolution of bankruptcy prediction models, from traditional financial ratio-based approaches to more advanced techniques such as machine learning and artificial intelligence. Researchers should continue to explore novel methodologies, incorporating both financial and non-financial variables, and enhance prediction accuracy and model performance (Pérez-Pons et al., 2022; Safi et al., 2022; Shetty et al., 2022; Lombardo et al., 2022; Papíková & Papík, 2022; Alam et al., 2021; Al-Milli et al., 2021; Jabeur et al., 2021; Hu et al., 2021; Perboli & Arabnezhad, 2021; Ghatasheh et al., 2020; Carmona et al., 2019; Climent et al., 2019; Hosaka, 2019). The use of ensemble methods, such as bagging and boosting, exhibits considerable potential and merits further investigation to determine its applicability in bankruptcy prediction across diverse industries and contexts (Khalil et al., 2022; Siswoyo et al., 2022; Tunio et al., 2021). However, Perboli et al. (2021) have shown that the introduction of potential false positives and negatives in predictions may limit the advantages of boosting compared to traditional methods, particularly when non-financial variables are incorporated. Furthermore, the exploration of qualitative information, incorporation of domain expertise and evaluation of the reasons behind bankruptcy can lead to a broader understanding of the financial distress (Alam et al., 2021).

Investors can benefit from the insights gained in this review by utilizing reliable bankruptcy prediction models to inform their investment decisions The identified critical financial ratios and indicators, such as liquidity, profitability, solvency, and return on investment, can serve as key factors to consider when assessing the creditworthiness and risk profile of potential investment opportunities. The use of simplified models / simple classifiers such as Support Vector Machines or Neural Networks that rely on easily obtainable financial ratios can be particularly valuable for investors who need to make timely decisions without extensive data collection. By employing accurate bankruptcy prediction models, investors can mitigate risks and make more informed investment choices (Shetty et al., 2022).

Policymakers can tap into the findings of this review to mitigate the fallouts from financial crises. By understanding the factors that contribute to bankruptcy, policymakers can proactively implement measures to support struggling organizations, implement regulatory frameworks and stimulate economic growth (Safi et al., 2022). Furthermore, the integration of external factors (i.e., economic conditions and industry-specific trends) into bankruptcy prediction models can enhance their effectiveness in capturing real-world dynamics and improving policy decisions (Perboli & Arabnezhad, 2021; Lakshmi et al., 2016).

Other stakeholders, such as creditors, lenders, and business owners, can also benefit from the insights of this review. By leveraging reliable bankruptcy prediction models, creditors and lenders can make informed decisions regarding loan approvals and risk assessments. Finally, business owners can avail of these models to monitor and manage their financial health, identify potential areas of improvement, and take proactive measures to avoid financial distress (Carmona et al., 2019; Climent et al., 2019).

5. Conclusion

This study reviewed 207 empirical studies on bankruptcy prediction between 2012 and mid-2023 using machine learning and artificial intelligence techniques. Our extensive review on bankruptcy prediction models highlighted the current trend toward the development of hybrid models that combine multiple methodologies and variables to improve bankruptcy prediction accuracy. Moreover, the peruse of the extant literature unveiled the problem of imbalanced datasets in bankruptcy prediction and the efforts made by researchers to address this problem by exploring and developing specialized techniques for handling imbalanced datasets. Another notable problem in bankruptcy prediction was overfitting. To overcome this problem, researchers used different metrics (i.e., sensitivity, specificity, Type I error, Type II error and Area Under the Curve) and approaches (i.e., splitting the dataset into training, validation, and test sets).

This study emphasized the significance of accurate bankruptcy prediction models in informing decision-making processes for academics, investors, policymakers, and other stakeholders. By integrating the identified methodologies and variables, researchers will be able to develop novel and more effective hybrid bankruptcy prediction models. Ultimately, these models will enhance the ability to identify and mitigate financial risks, support investment decisions, and foster the long-term stability and sustainability of organizations. With the development of hybrid models, predictive power will be enhanced, the risk of overfitting will be mitigated, and the models will offer improved interpretability compared to single models.

The limitation of our study is that it primarily focuses on hybrid models, with limited exploration of ensemble models.

Future research should be directed to expanding the application of bankruptcy prediction models to different datasets, industries, and economic conditions. As previously noted, numerous studies have relied on standard public datasets, leading to a degree of homogeneity in the data used. Furthermore, a notable gap exists in the examination of the banking sector, calling for further investigation. Given the distinct regulatory landscape governing banks, a more nuanced investigation of bankruptcy prediction could yield valuable insights. The inclusion of qualitative information, such as management expertise, textual information and strategic variables, could be further explored in order to enhance prediction accuracy.

As stated earlier, Principal Component Analysis (PCA), ANOVA tests, t-statistics and Factor Analysis have been the most commonly employed methods for feature selection. However, more advanced methods regarding the importance of variables have not yet been explored, suggesting a potential direction for future research. Methods such as Shapley Additive Explanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME) and Anchors could be employed by future studies. Additionally, the development of new hybrid models focused on feature selection would be another promising avenue for future research.

So far, the accuracy of predictions has not increased significantly, and therefore, there is a need for more research on the variables that perform more accurate predictions. The employment of non-traditional variables presents a potential avenue for future research. In addition to these variables, regulatory compliance metrics or environmental sustainability indicators could also be considered. These dimensions offer additional layers of insight, potentially enhancing the predictive capacity of models and contributing to a more comprehensive understanding of factors influencing financial stability. Other variables which have not been explored are the brand reputation and image, social media sentiment analysis, customer satisfaction surveys, social responsibility initiatives, social responsibility initiatives and the ability to adapt to market changes. Finally, the development of feature selection methods tailored to imbalanced data and the implementation of statistical tests across the varying results of each model will contribute to the advancement of bankruptcy prediction research. Future researchers should deploy statistical tests on their different models to enhance the robustness of their results. Conducting statistical tests will help validate the effectiveness and reliability of models, providing a more rigorous assessment of their performance. This approach will contribute to the robustness of empirical findings, ensuring that the reported outcomes are not subject to chance or random variation. By subjecting models to statistical scrutiny, researchers will be able to gain greater confidence in the generalizability and replicability of their results, thereby strengthening the overall quality of their study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

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Table A7
Comprehensive overview of the 207 selected articles.

Authors	Firm category	Dataset size	Country	Time period	Input variables
Chen (2012a)	Firms	100	Taiwan	2000–2007	Financial / Non-Financial variables/ Macroeconomic variables
Chen (2012b)	Firms	300	Taiwan	1999–2010	Financial / Non-financial variables/ Macroeconomic variables
Divsalar et al. (2012)	Firms	136	Iran	1999–2006	Financial variables
Du Jardin et al. (2012)	Firms	N/A	France	1991–2009	Financial variables
Huang et al (2012)	Firms	150	Taiwan	2000–2007	Financial variables
Jeong et al. (2012)	Firms	2,542	Korea	2001–2004	Financial variables
Li & Sun (2012)	Tourism Industry	23	China	1995–2010	Financial variables
Lin et al. (2012)	Firms	80	Taiwan	2000–2008	Financial variables
Shie et al. (2012)	Banks	54	USA	2000–2006	Financial variables
Chuang et al. (2013)	Firms	321	Taiwan	1999–2006	Financial variables
Lin & Chou (2013a)	Firms and Credit scoring	688 Chinese, 440 Taiwan, 690 Australian, 1,000 German	China, Taiwan, Australia,German	N/A	Financial variables
Lin & Chou (2013b)	Firms and Credit scoring	688 Chinese, 440 Taiwan, 690 Australian, 1,000 German	China, Taiwan, Australia, German	N/A	Financial variables
Arieshanti et al. (2013)	Firms	120	N/A	2 to 5 years	Financial variables
Chang et al. (2013)	Firms	120	Taiwan	1999–2009	Financial variables
Ecer (2013)	Commercial banks	34	Turkey	1994–2001	Financial variables
Erdal & Ekinci (2013)	Commercial banks	37	Turkey	1997–2001	Financial variables
Erdogan (2013)	Commercial banks	42	Turkey	1997–2003	Financial variables
Fedorova et al. (2013)	Firms	3,173	Russia	2007–2011	Financial variables
Garcia-Almanza et al. (2013)	Firms	N/A	USA	N/A	Financial variables
Korol (2013)	Firms	185 Polish dataset, 60 Latin America	Poland, Latin America	2000–2007, 1996–2009	Financial variables
Li & Sun (2013)	Firms	313	China	3-year period	Financial variables
Lin et al. (2013)	Firms	120,418	Korea	N/A	Financial variables
Tsai & Hsu (2013)	Firms	609 Australian, 1,000 German, 653 Japanese, 440 Tawain	Australia,German, Japan	1981–2009	Financial / Non-Financial variables
Zhang et al. (2013)	Firms	1	N/A	2006–2009	Financial variables
Zhou (2013)	Firms	N/A, 36,578 Japanese	USA, Japan	1981–2009, 1989–2009	Financial variables
Abellán & Mantas (2014)	Firms and Credit scoring	690 Japanese, 690 Australian, 1,000 German	Australia, German, Japan	N/A	Financial / Non-Financial variables
Chen (2014)	Firms	68	Taiwan	1994–2018	Financial / Non-Financial variables
Heo & Yang (2014)	Construction firms	29.862	Korea	2008–2012	Financial variables
Li & Jiulin (2014)	Real estate firms	67	China	2007–2013	Financial variables
Tsai (2014)	Firms and Credit scoring	690 Australian, 1,000 German, 653 Japanese, 241 Unknown	Australia,German, Japan	N/A	Financial / Non-Financial variables
Tsai et al. (2014)	Firms and Credit scoring	609 Australian, 1,000 German, 653 Japanese, 241 Unknown	Australia,German, Japan	N/A	Financial / Non-Financial variables
Wang et al. (2014)	Firms	240, 132	N/A	1997–2001, 1970–1982	Financial variables
Yu et al (2014)	Firms	1,020	France	2002–2003	Financial / Non-Financial variables
Zhou et al. (2014)	Firms	7,728	USA	1980–2006	Financial variables
Kim et al. (2015a)	Manufacturing firms	3,000	Korea	2002–2005	N/A
Kim et al. (2015b)	Manufacturing firms	3,000	Korea	2001–2005	Financial variables

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Table A7 (*continued*)

Authors	Firm category	Dataset size	Country	Time period	Input variables
Cheng & Hoan (2015)	Firms	76	N/A	1970–2011	Financial variables
Cheng et al. (2015)	Firms	76	N/A	1970–2011	Financial variables
Di Donato & Nieddu (2015)	Firms	100	Italy	2000–2011	Financial variables
Khademolqorani et al. (2015)	Manufactoring firms	180	Iran	2011	Financial variables
Liang et al. (2015)	Firms	688	China, Taiwan, Australia, German	2014	Financial / Non-Financial variables
López Iturriaga & Sanz (2015)	Firms	104	USA	2002–2012	Financial / Non-Financial variables
Lu, Zeng, et al. (2015)	Firms	250	N/A	N/A	Financial variables
Chung et al. (2016)	Firms	360	N/A	1998–2014	Financial variables
du Jardin (2016)	Firms	11 French Samples	France	2002–2012	Financial variables
Kim et al. (2016)	Firms	22,500	Korea	2002–2007	Financial variables
Klepac & Hampel (2016)	Firms	850	EU	2009–2013	Financial variables
Lakshmi et al. (2016)	Banks	N/A	N/A	N/A	Financial variables
Liang et al. (2016)	Firms	478	Taiwan	1999–2009	Financial + corporate
Sartori et al. (2016)	Firms	1,160	Italy	2013	Financial / Non-Financial variables
Slavici et al. (2016)	Firms	55	EU	1994–1998	Financial variables
Zhu (2016)	Firms	206	N/A	N/A	Financial variables
Zięba et al. (2016)	Firms	10,700	Poland	2000–2012	Financial / Non-Financial variables/ Macroeconomic variables
Antunes et al. (2017)	Firms and Credit scoring	2,000 French, 609 Australian, 1,000 German, 653 Japanese	France, Australia, German,Japan,	2002–2006	Financial variables
Azadnia et al. (2017)	N/A	N/A	France	N/A	N/A
Barboza et al. (2017)	Firms	N/A	N/A	1985–2006	Financial variables
Chou et al. (2017)	Firms	600	Taiwan	N/A	Financial variables
Ekinci & Erdal (2017)	Banks	37	Turkey	1998–2001	Financial variables
Fallahpour et al. (2017)	Firms	180	Tehran	1996–2014	Financial variables
Huang et al. (2017)	Firms	N/A	China	2000–2008	Financial variables
Jones et al. (2017)	Firms	30,129	USA	2007–2010	Financial variables
Karas & Reznakova (2017)	Firms	1,540	EU	2011–2014	Financial variables
Klepáč & Hampel (2017)	Firms	250	EU	2009–2013	Financial / Non-Financial variables
Tian & Yue (2017)	Firms	40	China	2017–2019	Financial variables
Wang & Wu (2017)	Firms	260	China	2012–2014	Financial variables
Zhou & Lai (2017)	Firms	2,236	USA	2007–2009	Financial variables
Cheng et al. (2018)	Firms	13,452	Taiwan	2008–2013	Financial variables
Alaka et al. (2018)	Construction firms	30	N/A	2001–2015	Financial variables
Andreano et al. (2018)	Manufactoring firms	2,643	Italy	2008–2012	Financial variables
du Jardin (2018)	Firms	6,120	France	2006–2014	Financial variables
Halteh et al. (2018)	Banks	101	Islam	2014	Financial / Non-Financial variables
Kim et al. (2018)	Firms	288	Korea	2000–2013	Financial variables
Klepáč & Hampel (2018)	Manufactoring firms	1,000	EU	2014	Financial variables
Le, Lee, et al. (2018)	Firms	120,418	Korea	2016–2017	Financial variables
Le, Son, et al. (2018)	Firms	120,418	Korea	2016–2017	Financial variables
Liang et al. (2018)	Firms and Credit scoring	690 Australian, 1,000 German, 688 Chinese, 440 Taiwanese	Australia,German, China, Tawain	N/A	Financial variables
Sisodia & Verma (2018)	Firms	N/A	Spain	N/A	Financial / Non-Financial variables
Staňková & Hampel (2018)	Firms	903	EU	2011–2013	Financial / Non-Financial variables
Affes & Hentati-Kaffel (2019)	Banks	1,247	USA	2008–2013	Financial variables
Azayite & Achchab (2019)	Firms	792	Morocco	N/A	Financial variables
Carmona et al. (2019)	Banks	156	USA	2001–2015	Financial variables
Climent et al. (2019)	Banks	168	EU	2006–2013	Financial variables
Devi & Radhika (2019)	Firms	255	N/A	N/A	Financial variables
Durica et al. (2019)	Firms	29	Poland	N/A	Financial variables
Farooq & Qamar (2019)	Firms	519	Pakistan	2002–2015	Financial / Non-Financial variables
Guha & Veeranjaneyulu (2019)	Firms	N/A	N/A	1994–2014	Financial variables
Hosaka (2019)	Firms	2,450	Japan	2002–2015	Financial / Non-Financial variables
Jang et al. (2019)	Firms	1,377	USA	1980–2016	Financial variables
Jang et al. (2019)	Firms	1,377	USA	1980–2016	Financial variables

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Authors	Firm category	Dataset size	Country	Time period	Input variables
Karminsky & Burekhin (2019)	Firms	3,981	Russia	2011–2017	Financial variables
Le, Vo, Fujita, et al. (2019)	Firms	120,418 Korean, 2,336 USA, 152 Japanese	Korea, USA, Japan	N/A, 1981–2009, 1995–2009	Financial / Non-Financial variables/ Macroeconomic variables
Le, Vo, Vo, et al. (2019)	Firms	N/A	Korea	2016–2017	Financial variables
Liu & Wu (2019)	Firms	920	China	2001–2014	Financial variables
Mai et al. (2019)	Firms	11,827	USA	1994–2014	Audit Reports
Muñoz-Izquierdo et al. (2019)	Firms	808	Spain	2004–2014	Financial variables,
Nguyen et al. (2019)	Firms	78,747	North America	N/A	Financial variables
Nyitrai & Virág (2019)	Firms	2,996	Hungary	2001–2016	Financial / Non-Financial variables
Pawełek (2019)	Firms	43,405	Poland	2000–2013	Financial variables
Son et al. (2019)	Firms	97,794	Korea	2011–2016	Financial variables
Song & Peng (2019)	Firms and Credit scoring	43,405 Polish, 1,000 German	Poland, German	N/A	Financial variables
Tang et al. (2019)	Firms	250	N/A	N/A	Financial variables
Valencia et al. (2019)	Firms	2,922	Colombia	2012–2013	Financial variables
Marso & Merouani, (2020a)	Firms	N/A	Poland	N/A	Financial variables
Marso & Merouani, (2020b)	Firms	N/A	Poland	N/A	Financial variables
Ansari et al. (2020)	Firms	984	USA	N/A	Financial / Non-Financial variables
Appiahene et al. (2020)	Banks	444	Ghana	N/A	Financial variables
Bateni & Asghari (2020)	Firms	174	Iran	2006–2014	Financial variables
Becerra-Vicario et al. (2020)	Firms	460	Spain	2008–2017	Financial variables
De Bock et al. (2020)	Firms	21 datasets	N/A	N/A	Financial variables
Faris et al. (2020)	Firms	478	Spain	1998–2003	Financial / Non-Financial variables
Ghatasheh et al. (2020)	Firms	478	Spain	1998–2003	Financial / Non-Financial variables
Gregova et al. (2020)	Firms	65	Slovakia	2015–2018	Financial / Non-Financial variables
J et al. (2020)	Firms	65	Slovakia	2015–2018	Financial variables
Lahmiri et al. (2020)	Construction firms	1,379	USA	1980–2016	Financial variables
Lahmiri et al. (2020)	Firms	1,379	USA	1980–2016	Financial variables
Lee et al. (2020)	Firms	2,342	N/A	N/A	Financial variables
Manthoulis et al. (2020)	Banks	60,000	USA	2006–2015	Financial / Non-Financial variables
Paule-Vianez et al. (2020)	Banks	148	Spain	2012–2016	Financial / Non-Financial variables
Roumani et al. (2020)	Firms	408	USA	2005–2018	Financial variables
Shrivastav & Janaki Ramudu (2020)	Banks	59	India	2000–2017	Financial variables
Smiti & Soui (2020)	Firms	43,405	Poland	2002–2013	Financial variables
Soui et al. (2020)	Firms	2 Datasets	N/A, Poland	2002–2013	Financial variables
Tsai (2020)	Firms	690 Australian, 1,000 German, 690 Japanese, 43,405 Polish, 6,820 Taiwanese	Australia,German, Japan,Poland, Taiwan	N/A	Financial variables
Tumpach et al. (2020)	Firms	N/A	Slovakia	2014–2019	Financial variables
Uthayakumar, Metawa, et al. (2020)	Firms	3 Datasets (250, 43,405 and 240)	N/A, Poland, N/A	N/A	Financial variables
Uthayakumar, Vengattaraman, et al. (2020)	Firms	3 Datasets (250, 43,405 and 240)	N/A, Poland, N/A	N/A	Financial variables
Viswanathan et al. (2020)	Banks	44	India	2005–2017	Financial variables
Vochozka et al. (2020)	Firms	55	Czech Republic	2014–2018	Financial variables
Zoričák et al. (2020)	Firms	N/A	EU	2010–2016	Financial / Non-Financial variables
Alam et al. (2021a)	Firms	43,405	Poland	2009–2013	Financial variables
Alam et al. (2021b)	Firms	6,481	USA	2001–2018	Financial variables
du Jardin (2021a)	Firms	30	France	2000–2016	Financial variables
du Jardin (2021b)	Firms	35	France	2000–2017	Financial variables
du Jardin (2021c)	Firms	30	France	2000–2016	Financial variables
Jabeur et al. (2021)	Firms	N/A	France	2014–2016	Financial variables
Abdulla (2021)	Firms	244	Banglandesh	2015–2019	Financial variables
Al-Milli et al. (2021)	Firms	1	Poland	2009–2013	Financial / Non-Financial variables
Aljawazneh et al. (2021)	Firms	471 Spanish, 6,819 Taiwanese, 1,000 Polish	Spain, Taiwan, Poland	1998–2003, 1999–2009, 2007–2013	Financial variables
Antulov-Fantulin et al. (2021)	Firms	7,795	Italy	2009–2016	Financial variables

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Table A7 (*continued*)

Authors	Firm category	Dataset size	Country	Time period	Input variables
Chen et al. (2021)	Firms	100	Taiwan	2000–2007	Financial / Non-Financial variables
Eygi Erdogan et al. (2021)	Banks	41	Turkey	1998–2001	Financial variables
Hartini et al. (2021)	Banks	521	79 countries	1981–1999	Financial / Non-Financial variables
He & Fan (2021)	Firms	30,000, 55,596	N/A	N/A	Financial variables
Hu et al. (2021)	Firms	57	Taiwan	2000–2008	Financial variables
Jandaghi et al. (2021)	Firms	218	Iran	1992–2017	Financial variables
Jang et al. (2021)	Construction firms	1,378	USA	1980–2016	Financial / Macroeconomic variables
Jencova et al. (2021)	Firms	987	Slovakia	2017–2018	Financial variables
Kim et al. (2021)	Electrical and engineering firms	987	Slovakia	20,182,017	Financial / Non-Financial variables/ Macroeconomic variables
Kim & Upneja (2021)	Firms	2,747	N/A	1980–2017	Financial variables
Metawa et al. (2021)	Firms	N/A	N/A	N/A	Financial variables
Mishraz et al. (2021)	Banks	75	India	2014–2019	Financial variables
Muslim & Dasril (2021)	Firms	43,405	Poland	2000–2012	Financial variables
Park et al. (2021)	Firms	546,619	Korea	2009–2015	Financial variables
Perboli & Arabnezhad (2021)	Firms	3,000	Italy	2001–2018	Financial variables
Saladi & Yarlagadda (2021)	Firms	15,458	Poland	N/A	Financial / Non-Financial variables
Schalck & Yankol-Schalck (2021)	SMEs	579,892	France	2012–2018	Financial variables
Tsai et al. (2021)	Firms	6 Bankruptcy datasets, 3 credit scores	N/A	N/A	Financial / Non-Financial variables
Tunio et al. (2021)	Firms	217	Pakistan	N/A	Financial / Non-Financial variables
Wang & Liu (2021)	Firms	6,819	Taiwan	1999–2009	Financial variables
Yang et al. (2021)	Firms	43,405 Polish, 35,960 Chinese	Poland, China	N/A	Financial variables
Yıldırım et al. (2021)	Firms	1,016,315	Turkey	2010–2018	N/A
Zhang et al. (2021)	Firms	240 Polish, 152 Japanese	Poland, Japan	N/A, 1995–2009	Financial variables
Abid et al. (2022)	Firms	856	France	2006	Financial variables
Arora & Saurabh (2022)	Firms and Credit scoring	524	India	N/A	Financial variables
Bragoli et al. (2022)	Industrial sector	5,560	Italy	2007–2015	Financial / Non-Financial variables
Brenes et al. (2022)	Firms	550	N/A	N/A	Financial variables
Cao et al. (2022)	Industrial sector	32,334	USA	1961–2018	Financial variables
Elhoseny, Metawa, Sztano, et al. (2022a)	Firms	50 Analcat Dataset, 10,503 Polish, 6,819 Taiwanese	Poland, Taiwan, and analcat dataset	2007–2013,1999–2009,N/A	Financial variables
Elhoseny, Metawa, Sztano, et al. (2022b)	Firms	30,000	France	2005–2016	Financial / Non-Financial variables
Gavurova et al. (2022)	Firms	2,384	Slovakia	2018–2019	Financial variables
Hilal et al. (2022)	Firms	27,703	Poland	N/A	Financial variables
Khalil et al. (2022)	Firms	N/A	Egypt	1999–2019	Financial variables
Kuang et al. (2022)	Firms	816	China	2010–2019	Financial variables
Kumar et al. (2022)	Firms	3 Datasets (250, 43,405 and 240)	N/A, Poland, Poland	N/A	N/A
Li et al. (2022)	Firms	4,426	USA	2002–2016	Financial / Non-Financial variables
Ling & Cai (2022)	Manufacturing, Financial, and Service industries	20	N/A	2019–2021	Financial variables
Liu et al. (2022a)	Firms	N/A	119 countries	1970–2017	Financial / Non-Financial variables
Liu et al. (2022b)	Start-ups	N/A	EU	2005–2014	Financial variables
Lombardo et al. (2022)	Firms	8,262	USA	1999–2018	Financial variables
Maryam et al. (2022)	Firms	359	India	2021	Financial variables
Papíková & Papík (2022)	SMEs	89,851	Slovakia	2019	Financial variables
Park & Yang (2022)	Firms	N/A	G20 Countries	1990–2019	Financial / Macroeconomic variables
Pérez-Pons et al. (2022)	SMEs	89,851	Slovakia	2019	Financial variables
Qian et al. (2022)	Firms	4,167	China	2001–2019	Financial variables
Safi et al. (2022)	Firms	6,888	Spain	2016–2018	Financial variables
Shetty et al. (2022)	SMEs	3,728	Belgium	2002–2012	Financial variables
Siswoyo et al. (2022)	Banks	N/A	Islam	2010–2016	Financial variables
Smith & Alvarez (2022)	Firms	64,057	Spain	1992–2016	Financial variables
Vaiyapuri et al. (2022)	Firms	3 Datasets (250, 43,405 and 240)	N/A, Poland, Poland	N/A	Financial variables
Venkateswarlu et al. (2022)	N/A	690 Australian, 1,000 German	Australia, German	N/A	Financial / Non-Financial variables

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Table A7 (*continued*)

Authors	Firm category	Dataset size	Country	Time period	Input variables
Zhang et al. (2022)	Banks	N/A	Brazil	2012–2021	Financial/ Macroecnomic variables/ Textual informations
Zou et al. (2022)	Firms	3,058 Chinese, 6,819 Taiwanese	China, Taiwan	N/A	Financial variables
Borchert et al. (2023)	Firms	13,571	EU	N/A	Financial / Non-Financial variables
Adisa et al. (2023)	Firms	43,405	Poland	2000–2013	Financial variables
Aranha & Bolar (2023)	Firms	5,000	N/A	2010–2021	Financial variables
Ben Jabeur et al. (2023)	Firms	1,850	France	2009–2017	Financial variables
Chen et al. (2023)	Firms	41,439	USA	1994–2018	Financial variables
Cho & Shin (2023)	Firms	440	Taiwan	1999–2009	Financial variables
De Jesus & Besarria (2023)	Banks	N/A	Brazil	2012–2021	Financial/ Macroecnomic variables/ Textual informations
Du Jardin (2023)	Firms	65,760	France	N/A	Financial variables
Dube et al. (2023)	Manufactoring firms	37	South Africa	2000–2019	Financial variables/ Macroeconomic
Nießner et al. (2023)	Firms	N/A	German	2017–2021	Financial variables
Oberoi & Banerjee (2023)	Banks	59	India	2000–2018	Macroeconomic and market structure variables
Radovanovic & Haas (2023)	Firms	18,858	North America	1985–2020	Financial variables/ Socioeconomic
Ranjbaran et al. (2023)	Firms	83,028 Polish, 8,262 USA	Poland, USA	2000–2012, 1999–2018	Financial variables
Sue et al. (2023)	Firms	3,058 Chinese, 6,819 Taiwanese	China, Taiwan	N/A	Financial variables
Zhao et al. (2023)	Firms	120	N/A	2018–2021	Financial variables

Source: Author's compilation

References

Abdullah, M. (2021). The implication of machine learning for financial solvency prediction: An empirical analysis on public listed companies of Bangladesh. *Journal of Asian Business and Economic Studies, 28*(4), 303–320. https://doi.org/10.1108/JABES-11-2020-0128

Abellán, J., & Mantas, C. J. (2014). Improving experimental studies about ensembles of classifiers for bankruptcy prediction and credit scoring. *Expert Systems with Applications, 41*(8), 3825–3830. https://doi.org/10.1016/j.eswa.2013.12.003

Abid, I., Ayadi, R., Guesmi, K., & Mkaouar, F. (2022). A new approach to deal with variable selection in neural networks: An application to bankruptcy prediction. *Annals of Operations Research, 313*(2), 605–623. https://doi.org/10.1007/s10479-021-04236-4

Adisa, J. A., Ojo, S., Owolawi, P. A., Pretorius, A., & Ojo, S. O. (2023). Application of an improved optimization using learning strategies and long short term-memory for bankruptcy prediction. *IAENG International Journal of Computer Science, 50*(2). https://www.scopus.com/inward/record.uri?eid=2-s2.0-85163708795&partnerID=40&md5=ef727b8815ce8beab9bccf3c345feca6.

Affes, Z., & Hentati-Kaffel, R. (2019). Forecast bankruptcy using a blend of clustering and MARS model: Case of US banks. *Annals of Operations Research, 281*(1–2), 27–64. https://doi.org/10.1007/s10479-018-2845-8

Ahmed, S., Alshater, M. M., Ammari, A. E., & Hammami, H. (2022). Artificial intelligence and machine learning in finance: A bibliometric review. *Research in International Business and Finance, 61*. https://doi.org/10.1016/j.ribaf.2022.101646

Alaka, H. A., Oyedele, L. O., Owolabi, H. A., Bilal, M., Ajayi, S. O., & Akinade, O. O. (2018). A framework for big data analytics approach to failure prediction of construction firms. *Applied Computing and Informatics, 16*(1–2), 207–222. https://doi.org/10.1016/j.aci.2018.04.003

Alam, N., Gao, J., & Jones, S. (2021). Corporate failure prediction: An evaluation of deep learning vs discrete hazard models. *Journal of International Financial Markets, Institutions and Money, 75*. https://doi.org/10.1016/j.intfin.2021.101455

Alam, T. M., Shaukat, K., Mushtaq, M., Ali, Y., Khushi, M., Luo, S., & Wahab, A. (2021). Corporate bankruptcy prediction: An approach towards better corporate world. *Computer Journal, 64*(11), 1731–1746. https://doi.org/10.1093/comjnl/bxaa056

Aljawazneh, H., Mora, A. M., Garcia-Sanchez, P., & Castillo-Valdivieso, P. A. (2021). Comparing the performance of deep learning methods to predict companies' financial failure. *IEEE Access, 9*, 97010–97038. https://doi.org/10.1109/ACCESS.2021.3093461

Al-Milli, N., Hudaib, A., & Obeid, N. (2021). Population diversity control of genetic algorithm using a novel injection method for bankruptcy prediction problem. *Mathematics, 9*(8). https://doi.org/10.3390/math9080823

Andreano, M. S., Benedetti, R., Mazzitelli, A., & Piersimoni, F. (2018). Spatial autocorrelation and clusters in modelling corporate bankruptcy of manufacturing firms. *Economia e Politica Industriale, 45*(4), 475–491. https://doi.org/10.1007/s40812-018-0097-x

Ansari, A., Ahmad, I. S., Bakar, A. A., & Yaakub, M. R. (2020). A hybrid metaheuristic method in training artificial neural network for bankruptcy prediction. *IEEE Access, 8*, 176640–176650. https://doi.org/10.1109/ACCESS.2020.3026529

Antulov-Fantulin, N., Lagravinese, R., & Resce, G. (2021). Predicting bankruptcy of local government: A machine learning approach. *Journal of Economic Behavior and Organization, 183*, 681–699. https://doi.org/10.1016/j.jebo.2021.01.014

Antunes, F., Ribeiro, B., & Pereira, F. (2017). Probabilistic modeling and visualization for bankruptcy prediction. *Applied Soft Computing Journal, 60*, 831–843. https://doi.org/10.1016/j.asoc.2017.06.043

Appiahene, P., Missah, Y. M., & Najim, U. (2020). Predicting bank operational efficiency using machine learning algorithm: Comparative study of decision rree, random forest, and neural networks. *Advances in Fuzzy Systems, 2020*. https://doi.org/10.1155/2020/8581202

Aranha, M., & Bolar, K. (2023). Efficacies of artificial neural networks ushering improvement in the prediction of extant credit risk models. *Cogent Economics and Finance, 11*(1). https://doi.org/10.1080/23322039.2023.2210916

Arieshanti, I., Purwananto, Y., Ramadhani, A., Nuha, M. U., & Ulinnuha, N. (2013). Comparative study of bankruptcy prediction models. *Telkomnika, 11*(3), 591–596. https://doi.org/10.12928/telkomnika.v11i3.1143.

Arora, P., & Saurabh, S. (2022). Predicting distress: A post insolvency and bankruptcy code 2016 analysis. *Journal of Economics and Finance, 46*(3), 604–622. https://doi.org/10.1007/s12197-022-09582-y

Azadnia, A. H., Siahi, A., & Motameni, M. (2017). An adaptive fuzzy neural network model for bankruptcy prediction of listed companies on the Tehran stock exchange. *International Journal of Engineering, Transactions B: Applications, 30*(12), 1879–1884. https://doi.org/10.5829/ije.2017.30.12c.09

Azayite, F. Z., & Achchab, S. (2019). Particle swarm optimization and simulated annealing to design bankruptcy prediction neural network. *Journal of Theoretical and Applied Information Technology, 97*(23), 3462–3474. https://www.scopus.com/inward/record.uri?eid=2-s2.0-85079221792&partnerID=40&md5=1b7d5d872ad41f5588be13af17d2bbf4.

Barboza, F., Kimura, H., & Altman, E. (2017). Machine learning models and bankruptcy prediction. *Expert Systems with Applications, 83*, 405–417. https://doi.org/10.1016/j.eswa.2017.04.006

Bateni, L., & Asghari, F. (2020). Bankruptcy prediction using logit and genetic algorithm models: A comparative analysis. *Computational Economics, 55*(1), 335–348. https://doi.org/10.1007/s10614-016-9590-3

Becerra-Vicario, R., Alaminos, D., Aranda, E., & Fernández-Gámez, M. A. (2020). Deep recurrent convolutional neural network for bankruptcy prediction: A case of the restaurant industry. *Sustainability (Switzerland), 12*(12). https://doi.org/10.3390/su12125180

Ben Jabeur, S., Stef, N., & Carmona, P. (2023). Bankruptcy prediction using the XGBoost algorithm and variable importance feature engineering. *Computational Economics, 61*(2), 715–741. https://doi.org/10.1007/s10614-021-10227-1

Borchert, P., Coussement, K., De Caigny, A., & De Weerdt, J. (2023). Extending business failure prediction models with textual website content using deep learning. *European Journal of Operational Research, 306*(1), 348–357. https://doi.org/10.1016/j.ejor.2022.06.060

Bragoli, D., Ferretti, C., Ganugi, P., Marseguerra, G., Mezzogori, D., & Zammori, F. (2022). Machine-learning models for bankruptcy prediction: Do industrial variables matter? *Spatial Economic Analysis, 17*(2), 156–177. https://doi.org/10.1080/17421772.2021.1977377

Brenes, R. F., Johannssen, A., & Chukhrova, N. (2022). An intelligent bankruptcy prediction model using a multilayer perceptron. *Intelligent Systems with Applications, 16*. https://doi.org/10.1016/j.iswa.2022.200136

Cao, Y., Liu, X., Zhai, J., & Hua, S. (2022). A two-stage Bayesian network model for corporate bankruptcy prediction. *International Journal of Finance and Economics, 27*(1), 455–472. https://doi.org/10.1002/ijfe.2162

Carmona, P., Climent, F., & Momparler, A. (2019). Predicting failure in the U.S. banking sector: An extreme gradient boosting approach. *International Review of Economics and Finance, 61*, 304–323. https://doi.org/10.1016/j.iref.2018.03.008

Chang, P.-C., Huang, C.-H., & Tsai, C.-Y. (2013). A hybrid system by the integration of case-based reasoning with support vector machine for prediction of financial crisis. *International Journal of Innovative Computing, Information and Control, 9*(6), 2245–2260. https://www.scopus.com/inward/record.uri?eid=2-s2.0-84880064136&partnerID=40&md5=053356c438216a1b353c38d3be2bcec0.

Chen, M.-Y. (2012a). Comparing traditional statistics, decision tree classification and support vector machine techniques for financial bankruptcy prediction. *Intelligent Automation and Soft Computing, 18*(1), 65–73. https://doi.org/10.1080/10798587.2012.10643227

Chen, M.-Y. (2012b). Visualization and dynamic evaluation model of corporate financial structure with self-organizing map and support vector regression. *Applied Soft Computing Journal, 12*(8), 2274–2288. https://doi.org/10.1016/j.asoc.2012.03.046

Chen, M.-Y. (2014). Using a hybrid evolution approach to forecast financial failures for Taiwan-listed companies. *Quantitative Finance, 14*(6), 1047–1058. https://doi.org/10.1080/14697688.2011.618458

Chen, T.-K., Liao, H.-H., Chen, G.-D., Kang, W.-H., & Lin, Y.-C. (2023). Bankruptcy prediction using machine learning models with the text-based communicative value of annual reports. *Expert Systems with Applications, 233*. https://doi.org/10.1016/j.eswa.2023.120714

Chen, Y.-S., Lin, C.-K., Lo, C.-M., Chen, S.-F., & Liao, Q.-J. (2021). Comparable studies of financial bankruptcy prediction using advanced hybrid intelligent classification models to provide early warning in the electronics industry. *Mathematics, 9*(20). https://doi.org/10.3390/math9202622

Cheng, C.-H., Chan, C.-P., & Yang, J.-H. (2018). A Seasonal time-series model based on gene expression programming for predicting financial distress. *Computational Intelligence and Neuroscience, 2018*. https://doi.org/10.1155/2018/1067350

Cheng, M.-Y., & Hoang, N.-D. (2015). Evaluating contractor financial status using a hybrid fuzzy instance based classifier: Case study in the construction industry. *IEEE Transactions on Engineering Management, 62*(2), 184–192. https://doi.org/10.1109/TEM.2014.2384513

Cho, S. H., & Shin, K.-S. (2023). Feature-weighted counterfactual-based explanation for bankruptcy prediction. *Expert Systems with Applications, 216*. https://doi.org/10.1016/j.eswa.2022.119390

Chou, C.-H., Hsieh, S.-C., & Qiu, C.-J. (2017). Hybrid genetic algorithm and fuzzy clustering for bankruptcy prediction. *Applied Soft Computing Journal, 56*, 298–316. https://doi.org/10.1016/j.asoc.2017.03.014

Chuang, C.-L. (2013). Application of hybrid case-based reasoning for enhanced performance in bankruptcy prediction. *Information Sciences, 236*, 174–185. https://doi.org/10.1016/j.ins.2013.02.015

Chung, C.-C., Chen, T.-S., Lin, L.-H., Lin, Y.-C., & Lin, C.-M. (2016). Bankruptcy prediction using cerebellar model neural networks. *International Journal of Fuzzy Systems, 18*(2), 160–167. https://doi.org/10.1007/s40815-015-0121-5

Ciampi, F., Giannozzi, A., Marzi, G., & Altman, E. I. (2021). Rethinking SME default prediction: A systematic literature review and future perspectives. *Scientometrics, 126*(3), 2141–2188. https://doi.org/10.1007/s11192-020-03856-0

Climent, F., Momparler, A., & Carmona, P. (2019). Anticipating bank distress in the Eurozone: An Extreme Gradient Boosting approach. *Journal of Business Research, 101*, 885–896. https://doi.org/10.1016/j.jbusres.2018.11.015

De Bock, K. W., Coussement, K., & Lessmann, S. (2020). Cost-sensitive business failure prediction when misclassification costs are uncertain: A heterogeneous ensemble selection approach. *European Journal of Operational Research, 285*(2), 612–630. https://doi.org/10.1016/j.ejor.2020.01.052

de Jesus, D. P., & Besarria, C. D. N. (2023). Machine learning and sentiment analysis: Projecting bank insolvency risk. *Research in Economics, 77*(2), 226–238. https://doi.org/10.1016/j.rie.2023.03.001

Devi, S., & Radhika, Y. (2019). An analytical approach for bankruptcy prediction using big data and machine learning technique. *Journal of Theoretical and Applied Information Technology, 97*(6), 1823–1832. https://www.scopus.com/inward/record.uri?eid=2-s2.0-85065491107&partnerID=40&md5=b5a14885eec3d9b059de3fd9b17e9bfb.

Devi, S. S., & Radhika, Y. (2018). A survey on machine learning and statistical techniques in bankruptcy prediction. *International Journal of Machine Learning and Computing, 8*(2), 133–139. https://doi.org/10.18178/ijmlc.2018.8.2.676.

Di Donato, F., & Nieddu, L. (2015). The effects of performance ratios in predicting corporate bankruptcy: The Italian case. *Lecture Notes in Business Information Processing, 216*, 61–72. https://doi.org/10.1007/978-3-319-18533-0_6

Divsalar, M., Roodsaz, H., Vahdatinia, F., Norouzzadeh, G., & Behrooz, A. H. (2012). A robust data-mining approach to bankruptcy prediction. *Journal of Forecasting, 31*(6), 504–523. https://doi.org/10.1002/for.1232

Du Jardin, P. (2016). A two-stage classification technique for bankruptcy prediction. *European Journal of Operational Research, 254*(1), 236–252. https://doi.org/10.1016/j.ejor.2016.03.008

du Jardin, P. (2018). Failure pattern-based ensembles applied to bankruptcy forecasting. *Decision Support Systems, 107*, 64–77. https://doi.org/10.1016/j.dss.2018.01.003

du Jardin, P. (2021a). Dynamic self-organizing feature map-based models applied to bankruptcy prediction. *Decision Support Systems, 147*. https://doi.org/10.1016/j.dss.2021.113576

du Jardin, P. (2021b). Forecasting bankruptcy using biclustering and neural network-based ensembles. *Annals of Operations Research, 299*(1–2), 531–566. https://doi.org/10.1007/s10479-019-03283-2

du Jardin, P. (2023). Designing topological data to forecast bankruptcy using convolutional neural networks. *Annals of Operations Research, 325*(2), 1291–1332. https://doi.org/10.1007/s10479-022-04780-7

Du Jardin, P., & Séverin, E. (2012). Forecasting financial failure using a Kohonen map: A comparative study to improve model stability over time. *European Journal of Operational Research, 221*(2), 378–396. https://doi.org/10.1016/j.ejor.2012.04.006

Dube, F., Nzimande, N., & Muzindutsi, P.-F. (2023). Application of artificial neural networks in predicting financial distress in the JSE financial services and manufacturing companies. *Journal of Sustainable Finance and Investment, 13*(1), 723–743. https://doi.org/10.1080/20430795.2021.2017257

Durica, M., Frnda, J., & Svabova, L. (2019). Decision tree based model of business failure prediction for Polish companies. *Oeconomia Copernicana, 10*(3), 453–469. https://doi.org/10.24136/oc.2019.022.

Ecer, F. (2013). Comparing the bank failure prediction performance of neural networks and support vector machines: The Turkish case [Usporedba performansi neuronskih mreža pri predvidstrokeanju propasti banaka i strojeva s potpornim vektorima: Slučaj Turske]. *Ekonomska Istrazivanja, 26*(3), 81–98. https://doi.org/10.1080/1331677X.2013.11517623

Ekinci, A., & Erdal, H.İ. (2017). Forecasting bank failure: Base learners, ensembles and hybrid ensembles. *Computational Economics, 49*(4), 677–686. https://doi.org/10.1007/s10614-016-9623-y

Elhoseny, M., Metawa, N., & El-hasnony, I. M. (2022). A new metaheuristic optimization model for financial crisis prediction: Towards sustainable development. *Sustainable Computing: Informatics and Systems, 35*. https://doi.org/10.1016/j.suscom.2022.100778

Elhoseny, M., Metawa, N., Sztano, G., & El-hasnony, I. M. (2022). Deep learning-based model for financial distress prediction. *Annals of Operations Research*. https://doi.org/10.1007/s10479-022-04766-5

Erdal, H. I., & Ekinci, A. (2013). A comparison of various artificial intelligence methods in the prediction of bank failures. *Computational Economics, 42*(2), 199–215. https://doi.org/10.1007/s10614-012-9332-0

Erdogan, B. E. (2013). Prediction of bankruptcy using support vector machines: An application to bank bankruptcy. *Journal of Statistical Computation and Simulation, 83*(8), 1543–1555. https://doi.org/10.1080/00949655.2012.666550

Eygi Erdogan, B., Özöğür-Akyüz, S., & Karadayı Ataş, P. (2021). A novel approach for panel data: An ensemble of weighted functional margin SVM models. *Information Sciences, 557*, 373–381. https://doi.org/10.1016/j.ins.2019.02.045

Fallahpour, S., Lakvan, E. N., & Zadeh, M. H. (2017). Using an ensemble classifier based on sequential floating forward selection for financial distress prediction problem. *Journal of Retailing and Consumer Services, 34*, 159–167. https://doi.org/10.1016/j.jretconser.2016.10.002

Faris, H., Abukhurma, R., Almanaseer, W., Saadeh, M., Mora, A. M., Castillo, P. A., & Aljarah, I. (2020). Improving financial bankruptcy prediction in a highly imbalanced class distribution using oversampling and ensemble learning: A case from the Spanish market. *Progress in Artificial Intelligence, 9*(1), 31–53. https://doi.org/10.1007/s13748-019-00197-9

Farooq, U., & Qamar, M. A. J. (2019). Predicting multistage financial distress: Reflections on sampling, feature and model selection criteria. *Journal of Forecasting, 38*(7), 632–648. https://doi.org/10.1002/for.2588

Fedorova, E., Gilenko, E., & Dovzhenko, S. (2013). Bankruptcy prediction for Russian companies: Application of combined classifiers. *Expert Systems with Applications, 40*(18), 7285–7293. https://doi.org/10.1016/j.eswa.2013.07.032

García, V., Marqués, A. I., & Sánchez, J. S. (2019). Exploring the synergetic effects of sample types on the performance of ensembles for credit risk and corporate bankruptcy prediction. *Information Fusion, 47*, 88–101. https://doi.org/10.1016/j.inffus.2018.07.004

Garcia-Almanza, A. L., Alexandrova-Kabadjova, B., & Martinez-Jaramillo, S. (2013). Bankruptcy prediction for banks: An artificial intelligence approach to improve understandability. *Studies in Computational Intelligence, 427*, 633–656. https://doi.org/10.1007/978-3-642-29694-9_24

Gavurova, B., Jencova, S., Bacik, R., Miskufova, M., & Letkovsky, S. (2022). Artificial intelligence in predicting the bankruptcy of non-financial corporations. *Oeconomia Copernicana, 13*(4), 1215–1251. https://doi.org/10.24136/oc.2022.035.

Ghatasheh, N., Faris, H., Abukhurma, R., Castillo, P. A., Al-Madi, N., Mora, A. M., Al-Zoubi, A. M., & Hassanat, A. (2020). Cost-sensitive ensemble methods for bankruptcy prediction in a highly imbalanced data distribution: A real case from the Spanish market. *Progress in Artificial Intelligence, 9*(4), 361–375. https://doi.org/10.1007/s13748-020-00219-x

Gregova, E., Valaskova, K., Adamko, P., Tumpach, M., & Jaros, J. (2020). Predicting financial distress of slovak enterprises: Comparison of selected traditional and learning algorithms methods. *Sustainability (Switzerland), 12*(10). https://doi.org/10.3390/SU12103954

Guha, A., & Veeranjaneyulu, N. (2019). Prediction of bankruptcy using big data analytic based on fuzzy C-means algorithm. *IAES International Journal of Artificial Intelligence, 8*(2), 168–174. https://doi.org/10.11591/ijai.v8.i2.pp168-174.

Halteh, K., Kumar, K., & Gepp, A. (2018). Financial distress prediction of Islamic banks using tree-based stochastic techniques. *Managerial Finance, 44*(6), 759–773. https://doi.org/10.1108/MF-12-2016-0372

Hartini, S., Rustam, Z., Saragih, G. S., & Vargas, M. J. S. (2021). Estimating probability of banking crises using random forest. *IAES International Journal of Artificial Intelligence, 10*(2), 407–413. https://doi.org/10.11591/IJAI.V10.I2.PP407-413.

He, H., & Fan, Y. (2021). A novel hybrid ensemble model based on tree-based method and deep learning method for default prediction. *Expert Systems with Applications, 176*. https://doi.org/10.1016/j.eswa.2021.114899

Heo, J., & Yang, J. Y. (2014). AdaBoost based bankruptcy forecasting of Korean construction companies. *Applied Soft Computing Journal, 24*, 494–499. https://doi.org/10.1016/j.asoc.2014.08.009

Hilal, A. M., Alsolai, H., Al-Wesabi, F. N., Al-Hagery, M. A., Hamza, M. A., & Duhayyim, M. A. (2022). Artificial intelligence based optimal functional link neural network for financial data Science. *Computers, Materials and Continua, 70*(3), 6289–6304. https://doi.org/10.32604/cmc.2022.021522.

Hosaka, T. (2019). Bankruptcy prediction using imaged financial ratios and convolutional neural networks. *Expert Systems with Applications, 117*, 287–299. https://doi.org/10.1016/j.eswa.2018.09.039

Hu, Y.-C., Jiang, P., Jiang, H., & Tsai, J.-F. (2021). Bankruptcy prediction using multivariate grey prediction models. *Grey Systems, 11*(1), 46–62. https://doi.org/10.1108/GS-12-2019-0067

Huang, J., Wang, H., & Kochenberger, G. (2017). Distressed Chinese firm prediction with discretized data. *Management Decision, 55*(5), 786–807. https://doi.org/10.1108/MD-08-2016-0546

Huang, S.-C., Tang, Y.-C., Lee, C.-W., & Chang, M.-J. (2012). Kernel local Fisher discriminant analysis based manifold-regularized SVM model for financial distress predictions. *Expert Systems with Applications, 39*(3), 3855–3861. https://doi.org/10.1016/j.eswa.2011.09.095

Jabeur, S. B., Gharib, C., Mefteh-Wali, S., & Arfi, W. B. (2021). CatBoost model and artificial intelligence techniques for corporate failure prediction. *Technological Forecasting and Social Change, 166*. https://doi.org/10.1016/j.techfore.2021.120658

Jandaghi, G., Saranj, A., Rajaei, R., Ghasemi, A., & Tehrani, R. (2021). Identification of the most critical factors in bankruptcy prediction and credit classification of companies. *Iranian Journal of Management Studies, 14*(4), 817–834. https://doi.org/10.22059/IJMS.2021.285398.673712.

Jang, Y., Jeong, I., & Cho, Y. K. (2021). Identifying impact of variables in deep learning models on bankruptcy prediction of construction contractors. *Engineering, Construction and Architectural Management, 28*(10), 3282–3298. https://doi.org/10.1108/ECAM-06-2020-0386

Jang, Y., Jeong, I.-B., Cho, Y. K., & Ahn, Y. (2019). Predicting business failure of construction contractors using long short-term memory recurrent neural network. *Journal of Construction Engineering and Management, 145*(11). https://doi.org/10.1061/(ASCE)CO.1943-7862.0001709

Jencova, S., Petruska, I., Lukacova, M., & Abu-Zaid, J. (2021). Prediction of bankruptcy in non-financial corporations using neural network. *Montenegrin Journal of Economics, 17*(4), 123–134. https://doi.org/10.14254/1800-5845/2021.17-4.11.

Jeong, C., Min, J. H., & Kim, M. S. (2012). A tuning method for the architecture of neural network models incorporating GAM and GA as applied to bankruptcy prediction. *Expert Systems with Applications, 39*(3), 3650–3658. https://doi.org/10.1016/j.eswa.2011.09.056

Jones, S., Johnstone, D., & Wilson, R. (2017). Predicting corporate bankruptcy: An evaluation of alternative statistical frameworks. *Journal of Business Finance and Accounting, 44*(1–2), 3–34. https://doi.org/10.1111/jbfa.12218

Karas, M., & Reznakova, M. (2017). Predicting the bankruptcy of construction companies: A CART-based model. *Engineering Economics, 28*(2), 145–154. https://doi.org/10.5755/j01.ee.28.2.16353

Karminsky, A. M., & Burekhin, R. N. (2019). Comparative analysis of methods for forecasting bankruptcies of Russian construction companies. *Business Informatics, 13*(3), 52–66. https://doi.org/10.17323/1998-0663.2019.3.52.66.

Khademolqorani, S., Zeinal Hamadani, A., & Mokhatab Rafiei, F. (2015). A hybrid analysis approach to improve financial distress forecasting: Empirical evidence from Iran. *Mathematical Problems in Engineering, 2015*. https://doi.org/10.1155/2015/178197

Khalil, A. A., Liu, Z., Salah, A., Fathalla, A., & Ali, A. (2022). Predicting insolvency of insurance companies in Egyptian market using bagging and boosting ensemble techniques. *IEEE Access, 10*, 117304–117314. https://doi.org/10.1109/ACCESS.2022.3210032

Kim, H., Cho, H., & Ryu, D. (2020). Corporate default predictions using machine learning: Literature review. *Sustainability (Switzerland), 12*(16). https://doi.org/10.3390/SU12166325

Kim, H., Cho, H., & Ryu, D. (2022). Corporate bankruptcy prediction using machine learning methodologies with a focus on sequential data. *Computational Economics, 59*(3), 1231–1249. https://doi.org/10.1007/s10614-021-10126-5

Kim, H.-J., Jo, N.-O., & Shin, K.-S. (2016). Optimization of cluster-based evolutionary undersampling for the artificial neural networks in corporate bankruptcy prediction. *Expert Systems with Applications, 59*, 226–234. https://doi.org/10.1016/j.eswa.2016.04.027

Kim, S., Mun, B. M., & Bae, S. J. (2018). Data depth based support vector machines for predicting corporate bankruptcy. *Applied Intelligence, 48*(3), 791–804. https://doi.org/10.1007/s10489-017-1011-3

Kim, S. Y., & Upneja, A. (2021). Majority voting ensemble with a decision trees for business failure prediction during economic downturns. *Journal of Innovation and Knowledge, 6*(2), 112–123. https://doi.org/10.1016/j.jik.2021.01.001

Kirkos, E. (2015). Assessing methodologies for intelligent bankruptcy prediction. *Artificial Intelligence Review, 43*(1), 83–123. https://doi.org/10.1007/s10462-012-9367-6

Klepac, V., & Hampel, D. (2017). Predicting financial distress of agriculture companies in EU. *Agricultural Economics (Czech Republic), 63*(8), 347–355. https://doi.org/10.17221/374/2015-AGRICECON.

Klepáč, V., & Hampel, D. (2018). Predicting bankruptcy of manufacturing companies in EU. *E a M: Ekonomie a Management, 21*(1), 159–174. https://doi.org/10.15240/tul/001/2018-1-011.

Korol, T. (2013). Early warning models against bankruptcy risk for Central European and Latin American enterprises. *Economic Modelling, 31*(1), 22–30. https://doi.org/10.1016/j.econmod.2012.11.017

Kuang, J., Chang, T.-C., & Chu, C.-W. (2022). Research on financial early warning based on combination forecasting model. *Sustainability (Switzerland), 14*(19). https://doi.org/10.3390/su141912046

Kuizinienė, D., Krilavičius, T., Damaševičius, R., & Maskeliūnas, R. (2022). Systematic review of financial distress identification using artificial intelligence methods. *Applied Artificial Intelligence, 36*(1). https://doi.org/10.1080/08839514.2022.2138124

Kumar, K. P. M., Dhanasekaran, S., Punithavathi, I. S. H., Duraipandy, P., Dutta, A. K., Pustokhina, I. V., & Pustokhin, D. A. (2022). Bird swarm algorithm with fuzzy min-max neural network for financial crisis prediction. *Computers, Materials and Continua, 73*(1), 1541–1555. https://doi.org/10.32604/cmc.2022.028338.

Lahmiri, S., Bekiros, S., Giakoumelou, A., & Bezzina, F. (2020). Performance assessment of ensemble learning systems in financial data classification. *Intelligent Systems in Accounting, Finance and Management, 27*(1), 3–9. https://doi.org/10.1002/isaf.1460

Lakshmi, T. M., Martin, A., & Venkatesan, V. P. (2016). A genetic bankrupt ratio analysis tool using a genetic algorithm to identify influencing financial ratios. *IEEE Transactions on Evolutionary Computation, 20*(1), 38–51. https://doi.org/10.1109/TEVC.2015.2424313

Lee, S., Choi, K., & Yoo, D. (2020). Predicting the insolvency of smes using technological feasibility assessment information and data mining techniques. *Sustainability (Switzerland), 12*(23), 1–17. https://doi.org/10.3390/su12239790

Li, H., & Sun, J. (2012). Forecasting business failure: The use of nearest-neighbour support vectors and correcting imbalanced samples - Evidence from the Chinese hotel industry. *Tourism Management, 33*(3), 622–634. https://doi.org/10.1016/j.tourman.2011.07.004

Li, H., & Sun, J. (2013). Predicting business failure using an RSF-based case-based reasoning ensemble forecasting method. *Journal of Forecasting, 32*(2), 180–192. https://doi.org/10.1002/for.1265

Li, Z., Feng, C., & Tang, Y. (2022). Bank efficiency and failure prediction: A nonparametric and dynamic model based on data envelopment analysis. *Annals of Operations Research, 315*(1), 279–315. https://doi.org/10.1007/s10479-022-04597-4

Liang, D., Lu, C.-C., Tsai, C.-F., & Shih, G.-A. (2016). Financial ratios and corporate governance indicators in bankruptcy prediction: A comprehensive study. *European Journal of Operational Research, 252*(2), 561–572. https://doi.org/10.1016/j.ejor.2016.01.012

Liang, D., Tsai, C.-F., Dai, A.-J., & Eberle, W. (2018). A novel classifier ensemble approach for financial distress prediction. *Knowledge and Information Systems, 54*(2), 437–462. https://doi.org/10.1007/s10115-017-1061-1

Liang, D., Tsai, C.-F., & Wu, H.-T. (2015). The effect of feature selection on financial distress prediction. *Knowledge-Based Systems, 73*(1), 289–297. https://doi.org/10.1016/j.knosys.2014.10.010

Lin, C.-T., & Chou, L.-D. (2013). A novel economy reflecting short-term load forecasting approach. *Energy Conversion and Management, 65*, 331–342. https://doi.org/10.1016/j.enconman.2012.08.001

Lin, F., Yeh, C.-C., & Lee, M.-Y. (2013). A hybrid business failure prediction model using locally linear embedding and support vector machines. *Romanian Journal of Economic Forecasting, 16*(1), 82–97. https://www.scopus.com/inward/record.uri?eid=2-s2.0-84878844306&partnerID=40&md5=08f863332e5d8a23694564c84feb528c.

Lin, W.-C., Lu, Y.-H., & Tsai, C.-F. (2019). Feature selection in single and ensemble learning-based bankruptcy prediction models. *Expert Systems, 36*(1). https://doi.org/10.1111/exsy.12335

Lin, W.-Y., Hu, Y.-H., & Tsai, C.-F. (2012). Machine learning in financial crisis prediction: A survey. *IEEE Transactions on Systems, Man and Cybernetics Part C: Applications and Reviews, 42*(4), 421–436. https://doi.org/10.1109/TSMCC.2011.2170420

Ling, T., & Cai, Y. (2022). Financial crisis prediction based on long-term and short-term memory meural network. *Wireless Communications and Mobile Computing, 2022*. https://doi.org/10.1155/2022/5728470

Liu, J., & Wu, C. (2019). Hybridizing kernel-based fuzzy c-means with hierarchical selective neural network ensemble model for business failure prediction. *Journal of Forecasting, 38*(2), 92–105. https://doi.org/10.1002/for.2561

Liu, L., Chen, C., & Wang, B. (2022). Predicting financial crises with machine learning methods. *Journal of Forecasting, 41*(5), 871–910. https://doi.org/10.1002/for.2840

Liu, Y., Zeng, Q., Li, B., Ma, L., & Ordieres-Meré, J. (2022). Anticipating financial distress of high-tech startups in the European Union: A machine learning approach for imbalanced samples. *Journal of Forecasting, 41*(6), 1131–1155. https://doi.org/10.1002/for.2852

Lombardo, G., Pellegrino, M., Adosoglou, G., Cagnoni, S., Pardalos, P. M., & Poggi, A. (2022). Machine learning for bankruptcy prediction in the American stock Market: Dataset and Benchmarks. *Future Internet, 14*(8). https://doi.org/10.3390/fi14080244

Lommers, K., El Harzli, O., & Kim, J. (2021). Confronting machine learning with financial research. *Journal of Financial Data Science, 3*(3), 67–96. https://doi.org/10.3905/jfds.2021.1.068

López Iturriaga, F. J., & Sanz, I. P. (2015). Bankruptcy visualization and prediction using neural networks: A study of U.S. commercial banks. *Expert Systems with Applications, 42*(6), 2857–2869. https://doi.org/10.1016/j.eswa.2014.11.025
Mai, F., Tian, S., Lee, C., & Ma, L. (2019). Deep learning models for bankruptcy prediction using textual disclosures. *European Journal of Operational Research, 274*(2), 743–758. https://doi.org/10.1016/j.ejor.2018.10.024
Manthoulis, G., Doumpos, M., Zopounidis, C., & Galariotis, E. (2020). An ordinal classification framework for bank failure prediction: Methodology and empirical evidence for US banks. *European Journal of Operational Research, 282*(2), 786–801. https://doi.org/10.1016/j.ejor.2019.09.040
Maryam, M., Anggoro, D. A., Tika, M. F., & Kusumawati, F. C. (2022). An intelligent hybrid model using artificial neural networks and particle swarm optimization technique for financial crisis prediction. *Pakistan Journal of Statistics and Operation Research, 18*(4), 1015–1025. https://doi.org/10.18187/pjsor.v18i4.3927.
Metawa, N., Nguyen, P. T., Nguyen, L. H. T. T., & Q., Elhoseny, M., & Shankar, K.. (2021). Internet of things enabled financial crisis prediction in enterprises using optimal feature subset selection-based classification model. *Big Data, 9*(5), 331–342. https://doi.org/10.1089/big.2020.0192
Mishraz, N., Ashok, S., & Tandon, D. (2021). Predicting financial distress in the Indian banking sector: A comparative study between the logistic regression, LDA and ANN Models. *Global Business Review*. https://doi.org/10.1177/09721509211026785.
Muñoz-Izquierdo, N., Camacho-Miñano, M.-M., Segovia-Vargas, M.-J., & Pascual-Ezama, D. (2019). Is the external audit report useful for bankruptcy prediction? Evidence using artificial intelligence. *International Journal of Financial Studies, 7*(2). https://doi.org/10.3390/ijfs7020020
Muslim, M. A., & Dasril, Y. (2021). Company bankruptcy prediction framework based on the most influential features using XGBoost and stacking ensemble learning. *International Journal of Electrical and Computer Engineering, 11*(6), 5549–5557. https://doi.org/10.11591/ijece.v11i6.pp5549-5557.
Nazareth, N., & Ramana Reddy, Y. V. (2023). Financial applications of machine learning: A literature review. *Expert Systems with Applications, 219*. https://doi.org/10.1016/j.eswa.2023.119640
Nguyen, S., Golas, E., Zywiak, W., & Kennedy, K. (2019). Dimension reduction in bankruptcy prediction: A case study of north american companies. *Advances in Business and Management Forecasting, 13*, 83–92. https://doi.org/10.1108/S1477-407020190000013010
Nießner, T., Nießner, S., & Schumann, M. (2023). Is it worth the effort? Considerations on text mining in ai-based corporate failure prediction. *Information (Switzerland), 14*(4). https://doi.org/10.3390/info14040215
Nyitrai, T., & Virág, M. (2019). The effects of handling outliers on the performance of bankruptcy prediction models. *Socio-Economic Planning Sciences, 67*, 34–42. https://doi.org/10.1016/j.seps.2018.08.004
Oberoi, S. S., & Banerjee, S. (2023). Bankruptcy prediction of indian banks using advanced analytics. *Ikonomicheski Izsledvania, 32*(4), 22–41. https://www.scopus.com/inward/record.uri?eid=2-s2.0-85159879914&partnerID=40&md5=228578e9a115f31afe76d5f2d8455299.
Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., McGuinness, L. A., Stewart, L. A., Thomas, J., Tricco, A. C., Welch, V. A., Whiting, P., & Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. In *The BMJ* (Vol. 372)BMJ Publishing Group. https://doi.org/10.1136/bmj.n71.
Papíková, L., & Papík, M. (2022). Effects of classification, feature selection, and resampling methods on bankruptcy prediction of small and medium-sized enterprises. *Intelligent Systems in Accounting, Finance and Management, 29*(4), 254–281. https://doi.org/10.1002/isaf.1521
Park, M. S., Son, H., Hyun, C., & Hwang, H. J. (2021). explainability of machine learning models for bankruptcy prediction. *IEEE Access, 9*, 124887–124899. https://doi.org/10.1109/ACCESS.2021.3110270
Park, S., & Yang, J.-S. (2022). Interpretable deep learning LSTM model for intelligent economic decision-making. *Knowledge-Based Systems, 248*. https://doi.org/10.1016/j.knosys.2022.108907
Paule-Vianez, J., Gutiérrez-Fernández, M., & Coca-Pérez, J. L. (2020). Prediction of financial distress in the Spanish banking system: An application using artificial neural networks. *Applied Economic Analysis, 28*(82), 69–87. https://doi.org/10.1108/AEA-10-2019-0039
Pawełek, B. (2019). Extreme gradient boosting method in the prediction of company bankruptcy. *Statistics in Transition New Series, 20*(2), 155–171. https://doi.org/10.21307/stattrans-2019-020.
Perboli, G., & Arabnezhad, E. (2021). A machine learning-based DSS for mid and long-term company crisis prediction. *Expert Systems with Applications, 174*. https://doi.org/10.1016/j.eswa.2021.114758
Pérez-Pons, M. E., Parra-Dominguez, J., Hernández, G., Herrera-Viedma, E., & Corchado, J. M. (2022). Evaluation metrics and dimensional reduction for binary classification algorithms: A case study on bankruptcy prediction. *Knowledge Engineering Review, 37*(4). https://doi.org/10.1017/S026988892100014X
Qian, H., Wang, B., Yuan, M., Gao, S., & Song, Y. (2022). Financial distress prediction using a corrected feature selection measure and radient Boosted Decision Tree. *Expert Systems with Applications, 190*, Article 116202. https://doi.org/10.1016/j.eswa.2021.116202
Radovanovic, J., & Haas, C. (2023). The evaluation of bankruptcy prediction models based on socio-economic costs. *Expert Systems with Applications, 227*. https://doi.org/10.1016/j.eswa.2023.120275
Ranjbaran, G., Reforgiato Recupero, D., Lombardo, G., & Consoli, S. (2023). Leveraging augmentation techniques for tasks with unbalancedness within the financial domain: A two-level ensemble approach. *EPJ Data Science, 12*(1). https://doi.org/10.1140/epjds/s13688-023-00402-9
Roumani, Y. F., Nwankpa, J. K., & Tanniru, M. (2020). Predicting firm failure in the software industry. *Artificial Intelligence Review, 53*(6), 4161–4182. https://doi.org/10.1007/s10462-019-09789-2
Safi, S.-A.-D., Castillo, P. A., & Faris, H. (2022). Cost-sensitive metaheuristic optimization-based neural network with ensemble learning for financial distress prediction. *Applied Sciences (Switzerland), 12*(14). https://doi.org/10.3390/app12146918
Saladi, S. D., & Yarlagadda, R. (2021). An enhanced bankruptcy prediction model using fuzzy clustering model and random forest algorithm. *Revue d'Intelligence Artificielle, 35*(1), 77–83. https://doi.org/10.18280/ria.350109.
Sartori, F., Mazzucchelli, A., & Gregorio, A. D. (2016). Bankruptcy forecasting using case-based reasoning: The CRePERIE approach. *Expert Systems with Applications, 64*, 400–411. https://doi.org/10.1016/j.eswa.2016.07.033
Schalck, C., & Yankol-Schalck, M. (2021). Predicting French SME failures: New evidence from machine learning techniques. *Applied Economics, 53*(51), 5948–5963. https://doi.org/10.1080/00036846.2021.1934389
Shetty, S., Musa, M., & Brédart, X. (2022). Bankruptcy prediction using machine learning techniques. *Journal of Risk and Financial Management, 15*(1). https://doi.org/10.3390/jrfm15010035
Shi, Y., & Li, X. (2019a). A bibliometric study on intelligent techniques of bankruptcy prediction for corporate firms. *Heliyon, 5*(12). https://doi.org/10.1016/j.heliyon.2019.e02997
Shi, Y., & Li, X. (2019b). An overview of bankruptcy prediction models for corporate firms: A systematic literature review. *Intangible Capital, 15*(2), 114–127. https://doi.org/10.3926/ic.1354
Shie, F. S., Chen, M.-Y., & Liu, Y.-S. (2012). Prediction of corporate financial distress: An application of the America banking industry. *Neural Computing and Applications, 21*(7), 1687–1696. https://doi.org/10.1007/s00521-011-0765-5
Shrivastav, S. K., & Janaki Ramudu, P. (2020). Bankruptcy prediction and stress quantification using support vector machine: Evidence from Indian banks. *Risks, 8*(2). https://doi.org/10.3390/risks8020052
Sisodia, D. S., & Verma, U. (2018). The impact of data re-sampling on learning performance of class imbalanced bankruptcy prediction models. *International Journal on Electrical Engineering and Informatics, 10*(3), 433–446. https://doi.org/10.15676/ijeei.2018.10.3.2.
Siswoyo, B., Abas, Z. A., Pee, A. N. C., Komalasari, R., & Suyatna, N. (2022). Ensemble machine learning algorithm optimization of bankruptcy prediction of bank. *IAES International Journal of Artificial Intelligence, 11*(2), 679–686. https://doi.org/10.11591/ijai.v11.i2.pp679-686.
Slavici, T., Maris, S., & Pirtea, M. (2016). Usage of artificial neural networks for optimal bankruptcy forecasting. Case study: Eastern European small manufacturing enterprises. *Quality and Quantity, 50*(1), 385–398. https://doi.org/10.1007/s11135-014-0154-0
Smith, M., & Alvarez, F. (2022). Predicting firm-level bankruptcy in the spanish economy using extreme gradient boosting. *Computational Economics, 59*(1), 263–295. https://doi.org/10.1007/s10614-020-10078-2
Smiti, S., & Soui, M. (2020). Bankruptcy prediction using deep learning approach based on borderline SMOTE. *Information Systems Frontiers, 22*(5), 1067–1083. https://doi.org/10.1007/s10796-020-10031-6
Son, H., Hyun, C., Phan, D., & Hwang, H. J. (2019). Data analytic approach for bankruptcy prediction. *Expert Systems with Applications, 138*. https://doi.org/10.1016/j.eswa.2019.07.033
Song, Y., & Peng, Y. (2019). A MCDM-Based evaluation approach for imbalanced classification methods in financial risk prediction. *IEEE Access, 7*, 84897–84906. https://doi.org/10.1109/ACCESS.2019.2924923
Soui, M., Smiti, S., Mkaouer, M. W., & Ejbali, R. (2020). Bankruptcy prediction using stacked auto-encoders. *Applied Artificial Intelligence, 34*(1), 80–100. https://doi.org/10.1080/08839514.2019.1691849
Staňková, M., & Hampel, D. (2018). Bankruptcy prediction of engineering companies in the EU using classification methods. *Acta Universitatis Agriculturae et Silviculturae Mendelianae Brunensis, 66*(5), 1347–1356. https://doi.org/10.11118/actaun201866051347.
Sue, K.-L., Tsai, C.-F., & Chiu, A. (2023). The data sampling effect on financial distress prediction by single and ensemble learning techniques. *Communications in Statistics - Theory and Methods, 52*(12), 4344–4355. https://doi.org/10.1080/03610926.2021.1992439
Tang, Y., Ji, J., Zhu, Y., Gao, S., Tang, Z., & Todo, Y. (2019). A differential evolution-oriented pruning neural network model for bankruptcy prediction. *Complexity, 2019*. https://doi.org/10.1155/2019/8682124
Tian, S., & Yue, H. (2017). Construction and research on enterprise financial crisis early warning based on artificial intelligence neural network. *Boletin Tecnico/Technical Bulletin, 55*(11), 19–27. https://www.scopus.com/inward/record.uri?eid=2-s2.0-85032910728&partnerID=40&md5=d0da7b1b90e44347f2c16a3dbc3b2988.
Tsai, C.-F. (2014). Combining cluster analysis with classifier ensembles to predict financial distress. *Information Fusion, 16*(1), 46–58. https://doi.org/10.1016/j.inffus.2011.12.001
Tsai, C.-F. (2020). Two-stage hybrid learning techniques for bankruptcy prediction*. *Statistical Analysis and Data Mining, 13*(6), 565–572. https://doi.org/10.1002/sam.11482
Tsai, C.-F., & Hsu, Y.-F. (2013). A meta-learning framework for bankruptcy prediction. *Journal of Forecasting, 32*(2), 167–179. https://doi.org/10.1002/for.1264
Tsai, C.-F., Hsu, Y.-F., & Yen, D. C. (2014). A comparative study of classifier ensembles for bankruptcy prediction. *Applied Soft Computing Journal, 24*, 977–984. https://doi.org/10.1016/j.asoc.2014.08.047

Tsai, C.-F., Sue, K.-L., Hu, Y.-H., & Chiu, A. (2021). Combining feature selection, instance selection, and ensemble classification techniques for improved financial distress prediction. *Journal of Business Research, 130*, 200–209. https://doi.org/10.1016/j.jbusres.2021.03.018

Tumpach, M., Surovičová, A., Juhászová, Z., Marci, A., & Kubaščíková, Z. (2020). Prediction of the bankruptcy of slovak companies using neural networks with SMOTE. *Ekonomicky Casopis, 68*(10), 1021–1039. https://doi.org/10.31577/ekoncas.2020.10.03.

Tunio, F. H., Ding, Y., Agha, A. N., Agha, K., & Panhwar, H. U. R. Z. (2021). Financial distress prediction using adaboost and bagging in pakistan stock exchange. *Journal of Asian Finance, Economics and Business, 8*(1), 665–673. https://doi.org/10.13106/jafeb.2021.vol8.no1.665.

Uthayakumar, J., Vengattaraman, T., & Dhavachelvan, P. (2020). Swarm intelligence based classification rule induction (CRI) framework for qualitative and quantitative approach: An application of bankruptcy prediction and credit risk analysis. *Journal of King Saud University - Computer and Information Sciences, 32*(6), 647–657. https://doi.org/10.1016/j.jksuci.2017.10.007

Vaiyapuri, T., Priyadarshini, K., Hemlathadhevi, A., Dhamodaran, M., Dutta, A. K., Pustokhina, I. V., & Pustokhin, D. A. (2022). Intelligent feature selection with deep learning based financial risk assessment model. *Computers, Materials and Continua, 72*(2), 2429–2444. https://doi.org/10.32604/cmc.2022.026204.

Valencia, C., Cabrales, S., Garcia, L., Ramirez, J., & Calderona, D. (2019). Generalized additive model with embedded variable selection for bankruptcy prediction: Prediction versus interpretation. *Cogent Economics and Finance, 7*(1). https://doi.org/10.1080/23322039.2019.1597956

Venkateswarlu, Y., Baskar, K., Wongchai, A., Gauri Shankar, V., Carranza, P. M., & C., Gonzáles, J. L. A., & Murali Dharan, A. R.. (2022). An efficient outlier detection with deep learning-based financial crisis prediction model in big data environment. *Computational Intelligence and Neuroscience, 2022*. https://doi.org/10.1155/2022/4948947

Viswanathan, P. K., Srinivasan, S., & Hariharan, N. (2020). Predicting financial health of banks for investor guidance using machine learning algorithms. *Journal of Emerging Market Finance, 19*(2), 226–261. https://doi.org/10.1177/0972652720913478

Vochozka, M., Vrbka, J., & Suler, P. (2020). Bankruptcy or success? The effective prediction of a company's financial development using LSTM. *Sustainability (Switzerland), 12*(18). https://doi.org/10.3390/su12187529

Wang, G., Ma, J., & Yang, S. (2014). An improved boosting based on feature selection for corporate bankruptcy prediction. *Expert Systems with Applications, 41*(5), 2353–2361. https://doi.org/10.1016/j.eswa.2013.09.033

Wang, H., & Liu, X. (2021). Undersampling bankruptcy prediction: Taiwan bankruptcy data. *PLoS ONE*, 16(7 July). https://doi.org/10.1371/journal.pone.0254030.

Wang, L., & Wu, C. (2017). Business failure prediction based on two-stage selective ensemble with manifold learning algorithm and kernel-based fuzzy self-organizing map. *Knowledge-Based Systems, 121*, 99–110. https://doi.org/10.1016/j.knosys.2017.01.016

Yang, D., Zhang, W., Wu, X., Ablanedo-Rosas, J. H., Yang, L., & Yu, W. (2021). A novel multi-stage ensemble model with fuzzy clustering and optimized classifier composition for corporate bankruptcy prediction. *Journal of Intelligent and Fuzzy Systems, 40*(3), 4169–4185. https://doi.org/10.3233/JIFS-200741

Yıldırım, M., Okay, F. Y., & Özdemir, S. (2021). Big data analytics for default prediction using graph theory. *Expert Systems with Applications, 176*. https://doi.org/10.1016/j.eswa.2021.114840

Yu, Q., Miche, Y., Séverin, E., & Lendasse, A. (2014). Bankruptcy prediction using Extreme Learning Machine and financial expertise. *Neurocomputing, 128*, 296–302. https://doi.org/10.1016/j.neucom.2013.01.063

Zhang, Y., Liu, R., Heidari, A. A., Wang, X., Chen, Y., Wang, M., & Chen, H. (2021). Towards augmented kernel extreme learning models for bankruptcy prediction: Algorithmic behavior and comprehensive analysis. *Neurocomputing, 430*, 185–212. https://doi.org/10.1016/j.neucom.2020.10.038

Zhang, Y., Wang, S., & Ji, G. (2013). A rule-based model for bankruptcy prediction based on an improved genetic ant colony algorithm. *Mathematical Problems in Engineering, 2013*. https://doi.org/10.1155/2013/753251

Zhang, Z., Luo, M., Hu, Z., & Niu, H. (2022). Textual emotional tone and financial crisis identification in chinese companies: A multi-source data analysis based on machine learning. *Applied Sciences (Switzerland), 12*(13). https://doi.org/10.3390/app12136662

Zhao, Z., Li, D., & Dai, W. (2023). Machine-learning-enabled intelligence computing for crisis management in small and medium-sized enterprises (SMEs). *Technological Forecasting and Social Change, 191*. https://doi.org/10.1016/j.techfore.2023.122492

Zhou, L. (2013). Performance of corporate bankruptcy prediction models on imbalanced dataset: The effect of sampling methods. *Knowledge-Based Systems, 41*, 16–25. https://doi.org/10.1016/j.knosys.2012.12.007

Zhou, L., & Lai, K. K. (2017). AdaBoost models for corporate bankruptcy prediction with missing data. *Computational Economics, 50*(1), 69–94. https://doi.org/10.1007/s10614-016-9581-4

Zhou, L., Lai, K. K., & Yen, J. (2014). Bankruptcy prediction using SVM models with a new approach to combine features selection and parameter optimisation. *International Journal of Systems Science, 45*(3), 241–253. https://doi.org/10.1080/00207721.2012.720293

Zhu, Q. (2016). Financial early warning dynamic evaluation model based on support vector machine. *Naukovyi Visnyk Natsionalnoho Hirnychoho Universytetu, 4*, 141–147. https://www.scopus.com/inward/record.uri?eid=2-s2.0-84991585559&partnerID=40&md5=5a59294ffff7415f0e030a3b1b653f02.

Zięba, M., Tomczak, S. K., & Tomczak, J. M. (2016). Ensemble boosted trees with synthetic features generation in application to bankruptcy prediction. *Expert Systems with Applications, 58*, 93–101. https://doi.org/10.1016/j.eswa.2016.04.001

Zoričák, M., Gnip, P., Drotár, P., & Gazda, V. (2020). Bankruptcy prediction for small- and medium-sized companies using severely imbalanced datasets. *Economic Modelling, 84*, 165–176. https://doi.org/10.1016/j.econmod.2019.04.003

Zou, Y., Gao, C., & Gao, H. (2022). Business failure prediction based on a cost-sensitive extreme gradient boosting machine. *IEEE Access, 10*, 42623–42639. https://doi.org/10.1109/ACCESS.2022.3168857